

# Mathematical Compendium for the Laplace AR(1) Diffusion Benchmark

*Full derivations, proofs, and interpretations accompanying the main memo*

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# Preface

This compendium accompanies the main memo “*Exact Benchmark for the Laplace AR(1) Diffusion*” and provides the full mathematical workings behind every statement made there. The main memo is designed to be read quickly; it states the results, shows the key formulas, and presents the numerical validation. It assumes the reader trusts the derivations. This document does not make that assumption. Every theorem is proved, every algebraic step is written out, every change of variables is justified, and every interpretation is anchored in the preceding algebra.

The style is deliberately pedagogical. When a step may look surprising, we pause to explain why it works — why a particular substitution is made, why a particular coordinate is natural, why a particular limit gives the behaviour we claim. This is meant to be a document one could hand to a collaborator (or to oneself six months from now) and recover the full mathematical story without having to reconstruct any step.

We chose a book format rather than a long article because the material divides naturally into six self-contained parts:

## **Part I — Setting and foundations.**

The generative model, the noising channel, and the structural constants that the whole analysis rests on.

## **Part II — The $K = 1$ structure theorems.**

Factorisation, residual density, joint score, curvature field. The core of the exact benchmark.

## **Part III — Fourier representation.**

The characteristic function, the rotation to innovation-frequency coordinates, the linear ODE satisfied by the residual density.

## **Part IV — Scalar observables.**

The Mills-ratio closed form for the centre curvature, Gaussian benchmark, kurtosis, FWHM.

## **Part V — The $K = 2$ case.**

The exact attempt, the obstruction to 1D factorisation, and the exact four-term representation.

## **Part VI — Interpretation.**

What Gaussianity buys us, what Markovianity buys us, and where the two structural roles can be disentangled.

Throughout, we keep the same notation as the original derivation note [1] (reproduced in Table 3.1 below) so that this document reads as a careful expansion of that note rather than a parallel account.

**How to read this document.** The main theorems are numbered and cross-referenced with the memo. Proofs are written out in full; each non-trivial algebraic step that we believe a reader could stumble on is boxed as a “why this step” note. An equation labelled (e.g.) **Eq. (5.15)** in the memo has the same reference here, so that one can flip between documents.

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## Part I

# Setting and foundations

# Chapter 1

## The generative model and the question we are asking

### 1.1 The clean chain

We study an autoregressive chain of order one driven by Laplace innovations. Fix parameters  $\alpha \in \mathbb{R}$ ,  $b > 0$ ,  $\sigma_0 > 0$ ,  $\mu_0 \in \mathbb{R}$ , and define

$$A_0 \sim \mathcal{N}(\mu_0, \sigma_0^2), \quad A_{k+1} = \alpha A_k + \varepsilon_{k+1}, \quad \varepsilon_k \stackrel{\text{iid}}{\sim} \text{Lap}(0, b), \quad (1.1)$$

where  $\text{Lap}(0, b)$  has density  $(2b)^{-1} \exp(-|u|/b)$ . The chain  $(A_k)_{k \geq 0}$  is a first-order Markov chain: conditional on  $A_k$ , the future  $A_{k+1}, A_{k+2}, \dots$  is independent of the past  $A_0, \dots, A_{k-1}$ . The innovation  $\varepsilon_k$  is the part of  $A_k$  that is new – not predictable from  $A_{k-1}$ .

We refer to  $(A_k)$  as the *clean chain*. The density of  $(A_0, \dots, A_{K-1})$  factorises:

$$p_0(a_0, \dots, a_{K-1}) = \phi_{\mu_0, \sigma_0^2}(a_0) \prod_{k=1}^{K-1} \frac{1}{2b} \exp\left(-\frac{|a_k - \alpha a_{k-1}|}{b}\right), \quad (1.2)$$

where  $\phi_{m, \sigma^2}$  denotes the Gaussian pdf with mean  $m$  and variance  $\sigma^2$ . This factorisation reflects the clean Markov property: the joint density is a product over bonds, each bond depending only on two consecutive variables.

**Why this step?** Equation (1.2) is what makes the AR(1) chain a Markov random field with a tridiagonal coupling structure. If we were to work in Fourier space, each bond would contribute a rational factor of the form  $1/(1 + b^2 \ell^2)$  to the characteristic function along its own innovation-frequency axis, a structure we will exploit heavily in Part III.

### 1.2 The OU corruption channel

Generative diffusion proceeds by smoothly noising the clean data through the Ornstein–Uhlenbeck (OU) channel:

$$X_k(t) = \beta A_k + \sqrt{\Delta} Z_k, \quad \beta := e^{-t}, \quad \Delta := 1 - e^{-2t}, \quad Z_k \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1), \quad (1.3)$$

with  $(Z_0, \dots, Z_{K-1})$  independent of the chain  $(A_0, \dots, A_{K-1})$ . For each fixed  $t$ , the transformation  $A_k \mapsto X_k(t)$  is affine-Gaussian and coordinatewise: each  $X_k(t)$  depends only on  $A_k$  and an independent Gaussian. In particular, conditional on  $A = (A_0, \dots, A_{K-1})$ , the coordinates  $X_0(t), \dots, X_{K-1}(t)$  are independent Gaussians.

This is the coordinate-local variant of the standard variance-preserving diffusion kernel; an alternative view starts from the SDE  $dX_k = -X_k dt + \sqrt{2} dB_k$  with initial condition  $X_k(0) = A_k$ , whose marginal distribution at time  $t$  is exactly (1.3).



**Why this step?** Three properties of the OU channel will be used over and over again: (i) at  $t = 0$  we have  $\beta = 1$ ,  $\Delta = 0$  and  $X = A$  identically; (ii) as  $t \rightarrow \infty$ ,  $\beta \rightarrow 0$ ,  $\Delta \rightarrow 1$  and  $X \rightarrow \mathcal{N}(0, I)$  regardless of the data; (iii) the channel is *conditionally independent across coordinates* given the clean state. Property (iii) is the reason Markov structure survives noising, as we will prove in Section 2.

### 1.3 The central question

Given the noisy distribution at time  $t$ ,  $p_t(x_0, x_1, \dots, x_{K-1})$ , we study the *joint score*

$$S(x, t) := \nabla_x \log p_t(x_0, \dots, x_{K-1}) \in \mathbb{R}^K, \quad x = (x_0, \dots, x_{K-1}), \quad (1.4)$$

and the *curvature field*

$$H_t(x) := -\nabla_x^2 \log p_t(x_0, \dots, x_{K-1}) \in \mathbb{R}^{K \times K}. \quad (1.5)$$

These are different from the per-frame marginal scores  $\nabla_{x_k} \log p_t(x_k)$ : the latter discard correlations between different time steps and therefore miss the whole point of diffusion over dynamic objects. Our question is:

*How does the joint score  $S(x, t)$  reflect the Markov structure of the clean chain, and what happens to that structure as  $t$  varies?*

In the Gaussian AR(1) benchmark (studied in [3]), the answer is known exactly:  $p_t$  is Gaussian,  $S$  is linear, and  $H_t = \Sigma_t^{-1}$  is a constant precision matrix that interpolates between tridiagonal (at  $t = 0$ , reflecting the Markov structure) and a full matrix (for large  $t$ ). The present compendium works out the answer in the simplest non-Gaussian case – the Laplace chain – where Gaussianity and Markovianity play separable structural roles. The resulting exact benchmark then serves as a fulcrum for the thesis: any general-purpose theory of joint scores over dynamic objects must recover this benchmark in its appropriate limits.

*Remark* (Joint vs. marginal scores). Throughout, “score” means joint score as in (1.4). Whenever we need the per-frame marginal score  $\nabla_{x_k} \log p_t(x_k)$  we will write “marginal score.”

## Chapter 2

# Why Markov structure survives noising

Before diving into explicit computations, we establish a structural fact that underpins all of Part II: the noisy law  $p_t$  is itself a Markov random field over time, with exactly the same sparsity graph as the clean law. This is not automatic – it requires the conditional independence property of the OU channel. The proof is short and instructive.

**Proposition 2.1** (Markov property survives OU noising). *Under (1.1)–(1.3), the noisy sequence  $(X_0, \dots, X_{K-1})$  is Markov of order one: for every  $k$ ,*

$$p_t(x_k \mid x_0, \dots, x_{k-1}) = p_t(x_k \mid x_{k-1}). \quad (2.1)$$

*Equivalently,  $X_{k+1} \perp (X_0, \dots, X_{k-1}) \mid X_k$ .*

*Proof.* We will use two facts repeatedly:

- (i) **Clean Markov property.**  $A_{k+1} \perp (A_0, \dots, A_{k-1}) \mid A_k$ .
- (ii) **Channel independence.** Given the entire clean chain  $A$ , the noises  $Z_0, \dots, Z_{K-1}$  are i.i.d.  $\mathcal{N}(0, 1)$  and independent of  $A$ , so  $(X_0, \dots, X_{K-1}) \mid A$  has independent coordinates and  $X_k \mid A = A_k$  depends only on  $A_k$ .

We compute  $p_t(x_k \mid x_0, \dots, x_{k-1})$  by conditioning on the clean chain:

$$p_t(x_k \mid x_0, \dots, x_{k-1}) = \int p_t(x_k \mid A, x_0, \dots, x_{k-1}) p(A \mid x_0, \dots, x_{k-1}) dA \quad (2.2)$$

$$= \int p_t(x_k \mid A_k) p(A \mid x_0, \dots, x_{k-1}) dA, \quad (2.3)$$

where the second equality uses channel independence (ii): given  $A$ ,  $X_k$  depends only on  $A_k$  and is independent of  $X_0, \dots, X_{k-1}$ . After integrating out  $A_{k+1}, \dots, A_{K-1}$  (which are irrelevant) and the components  $A_0, \dots, A_{k-1}$  (which do not appear in  $p_t(x_k \mid A_k)$ ), we are left with

$$p_t(x_k \mid x_0, \dots, x_{k-1}) = \int p_t(x_k \mid A_k) p(A_k \mid x_0, \dots, x_{k-1}) dA_k. \quad (2.4)$$

It remains to show that  $p(A_k \mid x_0, \dots, x_{k-1}) = p(A_k \mid x_{k-1})$ . Condition on  $A_{k-1}$  and use the clean Markov property (i):  $A_k \perp (A_0, \dots, A_{k-2}) \mid A_{k-1}$ . Channel independence gives  $A_k \perp (X_0, \dots, X_{k-2}) \mid A_{k-1}$ , and furthermore  $(A_k, A_{k-1}) \perp (X_0, \dots, X_{k-2}) \mid X_{k-1}$  because of the memoryless Markov structure of  $A_0 \rightarrow X_0, A_1 \rightarrow X_1$ , etc. Integrating out  $A_{k-1}$  then yields  $p(A_k \mid x_0, \dots, x_{k-1}) = p(A_k \mid x_{k-1})$ , so (2.4) reduces to

$$p_t(x_k \mid x_0, \dots, x_{k-1}) = \int p_t(x_k \mid A_k) p(A_k \mid x_{k-1}) dA_k = p_t(x_k \mid x_{k-1}),$$

which is (2.1). □

---

*Interpretation.* The noisy chain is genuinely Markov: every noisy transition  $p_t(x_{k+1} | x_k)$  encodes the full conditional dependence structure at time  $t$ . The deep reason is that the OU channel is *coordinate-local*: it never creates new coupling between coordinates at fixed  $t$ . Any coupling in  $(X_0, \dots, X_{K-1})$  must be inherited from the clean chain.

**Corollary 2.2** (Factorisation of the noisy joint density). *We have*

$$p_t(x_0, \dots, x_{K-1}) = p_t(x_0) \prod_{k=1}^{K-1} p_t(x_k | x_{k-1}). \quad (2.5)$$

*Remark* (Score structure). Because the joint log-density is a sum of two-point terms,  $\log p_t = \log p_t(x_0) + \sum_k \log p_t(x_k | x_{k-1})$ , the Hessian  $H_t = -\nabla^2 \log p_t$  is a sum of block matrices each of size  $2 \times 2$  supported on adjacent indices. This is why the joint score and curvature field have a tridiagonal block structure at  $K \geq 2$  even before we compute anything explicitly.

## Chapter 3

# Structural constants for $K = 1$

We now specialise to  $K = 1$ , i.e. we work with the pair  $(X_0, X_1)$  noised from  $(A_0, A_1)$ . This already contains the non-trivial physics: a Gaussian prior, a Laplace innovation, and a coordinate-local Gaussian channel. The goal of this chapter is to compute, from first principles, all the structural constants that appear in the exact formulas – the reparameterisations  $v$ ,  $s^2$ ,  $\tau^2$ ,  $\rho$ ,  $\nu$ ,  $\tilde{b}$ ,  $a$ . Every later derivation rests on these, so we are careful to show where each quantity comes from.

### 3.1 The marginal $p_t(x_0)$

Because  $X_0 = \beta A_0 + \sqrt{\Delta} Z_0$  with  $A_0 \sim \mathcal{N}(\mu_0, \sigma_0^2)$  and  $Z_0 \sim \mathcal{N}(0, 1)$  independent,

$$X_0 \sim \mathcal{N}(\beta\mu_0, v), \quad v := \beta^2\sigma_0^2 + \Delta, \quad (3.1)$$

and the density is

$$p_t(x_0) = \frac{1}{\sqrt{2\pi v}} \exp\left(-\frac{(x_0 - \beta\mu_0)^2}{2v}\right), \quad \partial_{x_0} \log p_t(x_0) = -\frac{x_0 - \beta\mu_0}{v}. \quad (3.2)$$

**Why this step?** The quantity  $v$  interpolates between  $\sigma_0^2$  (at  $t = 0$ , where  $\beta = 1, \Delta = 0$ ) and 1 (as  $t \rightarrow \infty$ , where  $\beta \rightarrow 0, \Delta \rightarrow 1$ ). It is the total variance of  $X_0$ , combining the prior variance (scaled down by  $\beta^2$ ) and the injected OU noise variance  $\Delta$ .

### 3.2 The posterior $A_0 \mid X_0 = x_0$

The joint  $(A_0, X_0)$  is bivariate Gaussian, so the posterior  $A_0 \mid X_0 = x_0$  is again Gaussian. The standard conditioning formulas are

$$m(x_0) := \mathbb{E}[A_0 \mid X_0 = x_0] = \mu_0 + \frac{\text{Cov}(A_0, X_0)}{\text{Var}(X_0)}(x_0 - \mathbb{E}X_0), \quad (3.3)$$

$$s^2 := \text{Var}(A_0 \mid X_0 = x_0) = \text{Var}(A_0) - \frac{\text{Cov}(A_0, X_0)^2}{\text{Var}(X_0)}. \quad (3.4)$$

We have  $\text{Cov}(A_0, X_0) = \text{Cov}(A_0, \beta A_0 + \sqrt{\Delta} Z_0) = \beta\sigma_0^2$ , so

$$m(x_0) = \mu_0 + \frac{\beta\sigma_0^2}{v}(x_0 - \beta\mu_0) = \frac{\Delta\mu_0 + \beta\sigma_0^2 x_0}{v}, \quad (3.5)$$

$$s^2 = \sigma_0^2 - \frac{\beta^2\sigma_0^4}{v} = \frac{\sigma_0^2\Delta}{v}. \quad (3.6)$$

**Why this step?** Equation (3.5) can be rewritten as  $m(x_0) = \mu_0 + \rho'(x_0 - \beta\mu_0)$  with  $\rho' = \beta\sigma_0^2/v$ ; this is the familiar “shrinkage factor” of Gaussian regression. Equation (3.6) says the posterior variance factorises into the prior variance  $\sigma_0^2$  times the ratio  $\Delta/v$ , which quantifies *how much the noise veils the signal*: it is zero at  $t = 0$  (we see  $A_0$  perfectly), and it approaches  $\sigma_0^2$  as  $t \rightarrow \infty$  (we see nothing).

### 3.3 The residual random variable

We now derive the key identity

$$X_1 \mid X_0 = x_0 \stackrel{d}{=} \nu + \rho x_0 + R_t, \quad R_t = \beta \varepsilon + G_t, \quad G_t \sim \mathcal{N}(0, \tau^2), \quad (3.7)$$

where  $R_t$  is independent of  $X_0$  and of the particular  $x_0$ . The constants  $\nu, \rho, \tau^2$  will be pinned down by direct computation.

Start from

$$X_1 = \beta A_1 + \sqrt{\Delta} Z_1 = \beta(\alpha A_0 + \varepsilon) + \sqrt{\Delta} Z_1 = \beta \alpha A_0 + \beta \varepsilon + \sqrt{\Delta} Z_1.$$

Decompose  $A_0 = m(x_0) + \xi$  where  $\xi := A_0 - m(x_0)$ . Conditional on  $X_0 = x_0$ ,  $\xi \sim \mathcal{N}(0, s^2)$  and, importantly,  $\xi$  is independent of  $(\varepsilon, Z_1)$ : indeed,  $\xi$  is a function of  $(A_0, X_0) = (A_0, \beta A_0 + \sqrt{\Delta} Z_0)$ , hence a linear combination of  $A_0$  and  $Z_0$ , both of which are independent of  $\varepsilon$  and  $Z_1$ . Therefore

$$X_1 \mid X_0 = x_0 \stackrel{d}{=} \beta \alpha m(x_0) + (\beta \alpha \xi + \beta \varepsilon + \sqrt{\Delta} Z_1). \quad (3.8)$$

The deterministic part simplifies using (3.5):

$$\beta \alpha m(x_0) = \beta \alpha \cdot \frac{\Delta \mu_0 + \beta \sigma_0^2 x_0}{v} = \underbrace{\frac{\alpha \beta \Delta \mu_0}{v}}_{=: \nu} + \underbrace{\frac{\alpha \beta^2 \sigma_0^2}{v}}_{=: \rho} x_0, \quad (3.9)$$

which defines the parameters  $\nu$  and  $\rho$ .

The random part of (3.8) is  $R_t := \beta \alpha \xi + \beta \varepsilon + \sqrt{\Delta} Z_1$ . Split it as

$$R_t = \beta \varepsilon + G_t, \quad G_t := \beta \alpha \xi + \sqrt{\Delta} Z_1.$$

The variable  $G_t$  is a linear combination of two independent Gaussians, hence Gaussian with mean zero and variance

$$\tau^2 := \text{Var}(G_t) = \beta^2 \alpha^2 s^2 + \Delta = \Delta + \alpha^2 \beta^2 \frac{\sigma_0^2 \Delta}{v} = \Delta \left( 1 + \frac{\alpha^2 \beta^2 \sigma_0^2}{v} \right). \quad (3.10)$$

Both  $\varepsilon$  and  $G_t$  are independent of each other:  $\varepsilon$  is independent of  $(\xi, Z_1)$  by construction and  $G_t$  is built from  $(\xi, Z_1)$ . Finally, the whole of  $R_t$  is independent of  $X_0$ : conditional on  $x_0$ , both  $\varepsilon$  (by independence) and  $G_t$  (with mean 0 and variance not depending on  $x_0$ ) are independent of the conditioning.

Collecting everything, we have proved (3.7) with

$$\boxed{\rho = \frac{\alpha \beta^2 \sigma_0^2}{v}, \quad \nu = \frac{\alpha \beta \Delta \mu_0}{v}, \quad \tau^2 = \Delta + \alpha^2 \beta^2 s^2, \quad s^2 = \frac{\sigma_0^2 \Delta}{v}.} \quad (3.11)$$

We also define  $\tilde{b} := \beta b$  – the effective Laplace scale of the *rescaled* innovation  $\beta \varepsilon$ , since  $\beta \varepsilon \sim \text{Lap}(0, \tilde{b})$ . The dimensionless quantity

$$a := \tau / \tilde{b} \quad (3.12)$$

will appear repeatedly and plays the role of an inverse non-Gaussianity parameter: large  $a$  means the Gaussian part of  $R_t$  dominates and  $R_t$  is nearly Gaussian; small  $a$  means the Laplace part dominates and the cusp at  $r = 0$  is strongly felt.

### 3.4 An auxiliary constant: the leakage $c$

A slightly different parameterisation will be useful in Chapters 8 and 16. Define the *innovation coordinate*

$$y := x_1 - \alpha x_0, \quad (3.13)$$

which is the noisy analogue of the clean innovation  $\varepsilon = A_1 - \alpha A_0$ . Direct substitution gives

$$Y := X_1 - \alpha X_0 = \beta(A_1 - \alpha A_0) + \sqrt{\Delta}(Z_1 - \alpha Z_0) = \beta\varepsilon + \sqrt{\Delta}(Z_1 - \alpha Z_0). \quad (3.14)$$

This variable contains no  $A_0$  contribution at all – its dependence on  $A$  is entirely through the innovation  $\varepsilon$ . But  $Y$  is not independent of  $X_0$  at finite  $t$ , because both contain the noise  $Z_0$ . To see how  $y$  and the residual  $r$  relate, compute

$$r = x_1 - \nu - \rho x_0 = y + \alpha x_0 - \nu - \rho x_0 = y + (\alpha - \rho)x_0 - \nu. \quad (3.15)$$

Using (3.11),

$$\alpha - \rho = \alpha - \frac{\alpha\beta^2\sigma_0^2}{v} = \alpha \cdot \frac{v - \beta^2\sigma_0^2}{v} = \alpha \cdot \frac{\Delta}{v} =: c, \quad (3.16)$$

so

$$r = y + c(x_0 - \beta\mu_0). \quad (3.17)$$

The constant  $c = \alpha\Delta/v$  measures how much the shared OU noise “leaks”  $x_0$ -dependence into the innovation channel. At  $t = 0$ ,  $\Delta = 0 \Rightarrow c = 0$ , and the residual and innovation coincide; at general  $t$ , a linear shift  $c(x_0 - \beta\mu_0)$  separates them.

| Symbol      | Definition                    | Meaning                    | Limit $t \rightarrow 0$ / $t \rightarrow \infty$ |
|-------------|-------------------------------|----------------------------|--------------------------------------------------|
| $\beta$     | $e^{-t}$                      | OU shrinkage               | 1 / 0                                            |
| $\Delta$    | $1 - \beta^2$                 | OU injected variance       | 0 / 1                                            |
| $v$         | $\beta^2\sigma_0^2 + \Delta$  | $\text{Var}(X_0)$          | $\sigma_0^2$ / 1                                 |
| $s^2$       | $\sigma_0^2\Delta/v$          | $\text{Var}(A_0   X_0)$    | 0 / $\sigma_0^2$                                 |
| $\rho$      | $\alpha\beta^2\sigma_0^2/v$   | residual slope in $x_0$    | $\alpha$ / 0                                     |
| $\nu$       | $\alpha\beta\Delta\mu_0/v$    | residual offset            | 0 / 0                                            |
| $c$         | $\alpha\Delta/v$              | innovation leakage         | 0 / $\alpha$                                     |
| $\tilde{b}$ | $\beta b$                     | rescaled Laplace scale     | $b$ / 0                                          |
| $\tau^2$    | $\Delta + \alpha^2\beta^2s^2$ | residual Gaussian variance | 0 / 1                                            |
| $a$         | $\tau/\tilde{b}$              | Mills argument             | 0 / $\infty$                                     |

Table 3.1: Structural constants and their limits. All derived from first principles in Chapter 3.

*Remark* (Check: the  $t \rightarrow 0$  limit). At  $t = 0$ :  $\beta = 1$ ,  $\Delta = 0$ ,  $v = \sigma_0^2$ ,  $s^2 = 0$ ,  $\tau = 0$ ,  $\rho = \alpha$ ,  $\nu = 0$ ,  $\tilde{b} = b$ ,  $a = 0$ . The residual becomes  $R_0 = \varepsilon$ , and (3.7) reduces to  $X_1 = \alpha X_0 + \varepsilon$ , i.e. the clean chain. Good.

*Remark* (Check: the  $t \rightarrow \infty$  limit). At  $t \rightarrow \infty$ :  $\beta \rightarrow 0$ ,  $\Delta \rightarrow 1$ ,  $v \rightarrow 1$ ,  $\tau^2 \rightarrow 1$ ,  $\rho \rightarrow 0$ ,  $\tilde{b} \rightarrow 0$ ,  $a \rightarrow \infty$ . The residual  $R_t = \beta\varepsilon + G_t$  becomes a standard normal, and the chain collapses to  $X \sim \mathcal{N}(0, I)$ , the diffusion prior. Good.

## Part II

### The $K=1$ structure theorems

## Chapter 4

# Factorisation theorem

The factorisation theorem states that the noisy joint density  $p_t(x_0, x_1)$  decomposes exactly as a product of a Gaussian site density and a one-dimensional non-Gaussian bond. Conceptually: the only non-Gaussian content of  $(X_0, X_1)$  lives in the one-dimensional residual coordinate  $r = x_1 - \nu - \rho x_0$ ; everything perpendicular to this direction is Gaussian.

**Theorem 4.1** (Factorisation,  $K=1$ ). *Under the model (1.1)–(1.3),*

$$p_t(x_0, x_1) = g_t(x_0) h_t(r), \quad r = x_1 - \nu - \rho x_0, \quad (4.1)$$

where  $g_t(x_0) = \frac{1}{\sqrt{2\pi v}} \exp(-(x_0 - \beta\mu_0)^2/(2v))$  is the Gaussian marginal of  $X_0$ , and  $h_t$  is the density of the Normal-Laplace random variable

$$R_t = \beta\varepsilon + G_t, \quad \varepsilon \sim \text{Lap}(0, b), \quad G_t \sim \mathcal{N}(0, \tau^2), \quad \varepsilon \perp G_t. \quad (4.2)$$

*Proof.* By the chain rule of probability,  $p_t(x_0, x_1) = p_t(x_0)p_t(x_1 | x_0)$ . We have already shown  $p_t(x_0) = g_t(x_0)$  in (3.2). From Section 3.3,

$$X_1 | X_0 = x_0 \stackrel{d}{=} \nu + \rho x_0 + R_t,$$

and  $R_t$  is independent of  $x_0$  (indeed, its distribution does not depend on  $x_0$  at all). By a standard change-of-variables argument, the conditional density is

$$p_t(x_1 | x_0) = h_t(x_1 - \nu - \rho x_0) = h_t(r). \quad (4.3)$$

Multiplying through gives (4.1). □

*Interpretation* (What this result is, and what it is not). Theorem 4.1 is *not* the clean factorisation of a Markov chain – that would read  $p_t(x_0, x_1) = p_t(x_0)p_t(x_1 | x_0)$ , which is always true and carries no information about the innovation structure. What (4.1) says is that the *conditional law*  $p_t(x_1 | x_0)$  is a *translate* of the same universal 1D density  $h_t$ , at an  $x_0$ -dependent location  $\nu + \rho x_0$ . This is a non-trivial consequence of the AR(1)+OU structure: both the deterministic shift  $\nu + \rho x_0$  and the shape  $h_t$  are explicit.

*Remark.* From this point on, every derivative of  $\log p_t(x_0, x_1)$  with respect to  $x_0$  or  $x_1$  can be computed via the chain rule through the single variable  $r$ . This is why the subsequent theorems all read as “take  $\psi_t := (\log h_t)'$ , and here are the scores/Hessians.”



## Chapter 5

# The Normal-Laplace residual density

The residual density  $h_t$  is the convolution of a Laplace and a Gaussian density. We compute it in closed form. The result is classical (it is the Normal-Laplace density used in financial modelling), but it is important to see the algebra because the *form* of the answer – two exponential terms, each multiplied by a Gaussian CDF – is what makes all subsequent derivations (score, curvature, Mills' ratio) tractable.

### 5.1 Setup

Since  $\beta\varepsilon \sim \text{Lap}(0, \tilde{b})$  with  $\tilde{b} = \beta b$ , the density of  $R_t = \beta\varepsilon + G_t$  with  $G_t \sim \mathcal{N}(0, \tau^2)$  independent is

$$h_t(r) = \int_{-\infty}^{\infty} \frac{1}{2\tilde{b}} e^{-|u|/\tilde{b}} \cdot \frac{1}{\sqrt{2\pi\tau^2}} e^{-(r-u)^2/(2\tau^2)} du. \quad (5.1)$$

The absolute value  $|u|$  prevents a direct Gaussian integration, so we split the real line at  $u = 0$ :

$$h_t(r) = \frac{1}{2\tilde{b}\sqrt{2\pi\tau^2}} [I_+(r) + I_-(r)], \quad (5.2)$$

with

$$I_+(r) := \int_0^{\infty} \exp\left(-\frac{u}{\tilde{b}} - \frac{(r-u)^2}{2\tau^2}\right) du, \quad I_-(r) := \int_{-\infty}^0 \exp\left(\frac{u}{\tilde{b}} - \frac{(r-u)^2}{2\tau^2}\right) du. \quad (5.3)$$

Each of  $I_{\pm}$  is a half-line Gaussian integral with a linear term – exactly the form that reduces to a complementary Gaussian CDF after completing the square.

### 5.2 Computation of $I_+(r)$

Expand the exponent of the  $I_+$  integrand:

$$-\frac{u}{\tilde{b}} - \frac{(r-u)^2}{2\tau^2} = -\frac{u}{\tilde{b}} - \frac{u^2 - 2ru + r^2}{2\tau^2} \quad (5.4)$$

$$= -\frac{u^2}{2\tau^2} + \left(\frac{r}{\tau^2} - \frac{1}{\tilde{b}}\right)u - \frac{r^2}{2\tau^2}. \quad (5.5)$$

We now complete the square in  $u$ . Write

$$\frac{r}{\tau^2} - \frac{1}{\tilde{b}} = \frac{r - \tau^2/\tilde{b}}{\tau^2}, \quad (5.6)$$

so that the linear coefficient in  $u$  equals  $(r - \tau^2/\tilde{b})/\tau^2$ . The quadratic-plus-linear combination becomes

$$-\frac{u^2}{2\tau^2} + \frac{r - \tau^2/\tilde{b}}{\tau^2} u = -\frac{1}{2\tau^2} \left[ u^2 - 2(r - \tau^2/\tilde{b}) u \right] \quad (5.7)$$

$$= -\frac{1}{2\tau^2} \left[ (u - r + \tau^2/\tilde{b})^2 - (r - \tau^2/\tilde{b})^2 \right]. \quad (5.8)$$

**Why this step?** Completing the square in  $u$  converts “Gaussian times exponential” into “Gaussian centred at a shifted point, times a deterministic factor.” The shifted centre  $u_* = r - \tau^2/\tilde{b}$  tells us where the integrand is concentrated; the deterministic factor  $(r - \tau^2/\tilde{b})^2 - r^2$  absorbs the parts that do not depend on  $u$ .

Combining the completion-of-squares with the  $-r^2/(2\tau^2)$  term from (5.5) gives

$$-\frac{u}{\tilde{b}} - \frac{(r - u)^2}{2\tau^2} = -\frac{(u - r + \tau^2/\tilde{b})^2}{2\tau^2} + \frac{(r - \tau^2/\tilde{b})^2 - r^2}{2\tau^2} \quad (5.9)$$

$$= -\frac{(u - r + \tau^2/\tilde{b})^2}{2\tau^2} - \frac{r}{\tilde{b}} + \frac{\tau^2}{2\tilde{b}^2}. \quad (5.10)$$

The last step uses  $\frac{(r - \tau^2/\tilde{b})^2 - r^2}{2\tau^2} = \frac{-2r\tau^2/\tilde{b} + \tau^4/\tilde{b}^2}{2\tau^2} = -\frac{r}{\tilde{b}} + \frac{\tau^2}{2\tilde{b}^2}$ .

Pulling the deterministic factor out of the integral,

$$I_+(r) = e^{-r/\tilde{b} + \tau^2/(2\tilde{b}^2)} \int_0^\infty \exp\left(-\frac{(u - r + \tau^2/\tilde{b})^2}{2\tau^2}\right) du. \quad (5.11)$$

Change variable:  $z = (u - r + \tau^2/\tilde{b})/\tau$ , so  $du = \tau dz$  and the lower limit  $u = 0$  becomes  $z = -r/\tau + \tau/\tilde{b}$ . Using  $a = \tau/\tilde{b}$ ,

$$I_+(r) = e^{-r/\tilde{b} + a^2/2} \tau \int_{-r/\tau + a}^\infty e^{-z^2/2} dz = e^{-r/\tilde{b} + a^2/2} \tau \sqrt{2\pi} \Phi\left(\frac{r}{\tau} - a\right). \quad (5.12)$$

Here  $\Phi(x) := \int_{-\infty}^x \phi(z) dz$  is the standard normal CDF, and we used  $\int_c^\infty e^{-z^2/2} dz = \sqrt{2\pi} \Phi(-c)$  together with  $\Phi(-c) = \Phi(c)|_{c \rightarrow -c}$ .

*Remark* (Check the exponent constants). The prefactor  $\exp(-r/\tilde{b} + a^2/2)$  has a natural interpretation:  $-r/\tilde{b}$  is the Laplace exponential decay in the tail  $r \rightarrow +\infty$ , and  $a^2/2 = \tau^2/(2\tilde{b}^2)$  is a finite correction coming from the Gaussian smoothing. As  $\tau \rightarrow 0$  the correction vanishes and we recover pure Laplace behaviour.

### 5.3 Computation of $I_-(r)$

By the symmetry  $u \mapsto -u$ ,

$$I_-(r) = \int_{-\infty}^0 \exp\left(\frac{u}{\tilde{b}} - \frac{(r - u)^2}{2\tau^2}\right) du = \int_0^\infty \exp\left(-\frac{u}{\tilde{b}} - \frac{(r + u)^2}{2\tau^2}\right) du. \quad (5.13)$$

This is the same integral as  $I_+$  with  $r \rightarrow -r$ . Therefore, by (5.12),

$$I_-(r) = e^{r/\tilde{b} + a^2/2} \tau \sqrt{2\pi} \Phi\left(-\frac{r}{\tau} - a\right). \quad (5.14)$$

## 5.4 The closed form of $h_t$

Substituting (5.12) and (5.14) into (5.2) and simplifying the constants:

$$\begin{aligned} h_t(r) &= \frac{1}{2\tilde{b}\sqrt{2\pi\tau^2}} \cdot e^{a^2/2} \tau\sqrt{2\pi} \left[ e^{-r/\tilde{b}} \Phi\left(\frac{r}{\tau} - a\right) + e^{r/\tilde{b}} \Phi\left(-\frac{r}{\tau} - a\right) \right] \\ &= \frac{e^{a^2/2}}{2\tilde{b}} \left[ e^{-r/\tilde{b}} \Phi\left(\frac{r}{\tau} - a\right) + e^{r/\tilde{b}} \Phi\left(-\frac{r}{\tau} - a\right) \right]. \end{aligned} \quad (5.15)$$

This is the Normal–Laplace density; it is the key formula of Part II.

*Interpretation.* The two exponentials capture the two Laplace tails (decay to the right and to the left), each damped by a Gaussian-CDF factor that smooths out the cusp at  $r = 0$ . In the limit  $\tau \rightarrow 0$  (so  $a \rightarrow 0$ ), each  $\Phi(\cdot - a)$  becomes a Heaviside step that selects the correct half of the Laplace, and we recover the clean Laplace density  $e^{-|r|/\tilde{b}}/(2\tilde{b})$  as expected.

*Remark* (Sanity check:  $h_t$  integrates to 1). As  $r \rightarrow \pm\infty$ , one of the two exponentials dominates and the other vanishes; the Gaussian CDF in the dominant branch approaches 1, recovering pure Laplace decay. Integrating  $e^{-|r|/\tilde{b}}/(2\tilde{b})$  over  $\mathbb{R}$  gives 1, which is the leading contribution. The finite- $\tau$  corrections cancel; a direct computation via the convolution definition confirms  $\int h_t(r) dr = 1$ .

**Definition 5.1** ( $L_{\pm}$  and the compact form). We introduce the notation

$$L_+(r) := e^{-r/\tilde{b}} \Phi\left(\frac{r}{\tau} - a\right), \quad L_-(r) := e^{r/\tilde{b}} \Phi\left(-\frac{r}{\tau} - a\right), \quad (5.16)$$

so that

$$h_t(r) = \frac{e^{a^2/2}}{2\tilde{b}} [L_+(r) + L_-(r)]. \quad (5.17)$$

## Chapter 6

# Score theorem

The next building block is the closed form of the joint score. We compute the residual score  $\psi_t = (\log h_t)'$  first; the full joint score then follows by the chain rule through  $r$ .

### 6.1 A useful identity

Before differentiating, we record an identity that makes the computation clean.

**Lemma 6.1** (Gaussian-derivative cancellation).

$$e^{-r/\tilde{b}} \phi\left(\frac{r}{\tau} - a\right) = e^{r/\tilde{b}} \phi\left(\frac{r}{\tau} + a\right) = e^{-a^2/2} \phi\left(\frac{r}{\tau}\right). \quad (6.1)$$

*Proof.* Using  $\phi(x) = (2\pi)^{-1/2} e^{-x^2/2}$  and  $(r/\tau - a)^2 = (r/\tau)^2 - 2ar/\tau + a^2$ , we get  $\phi(r/\tau - a) = \phi(r/\tau) e^{ar/\tau - a^2/2}$ . Since  $a/\tau = 1/\tilde{b}$ ,

$$e^{-r/\tilde{b}} \phi(r/\tau - a) = e^{-r/\tilde{b}} \phi(r/\tau) e^{r/\tilde{b} - a^2/2} = e^{-a^2/2} \phi(r/\tau). \quad (6.2)$$

The other equality is by the same argument with  $r \rightarrow -r$ .  $\square$

**Why this step?** This identity is at the heart of why  $h_t'$  simplifies. When we differentiate  $L_+ + L_-$ , each piece produces two terms: one from the exponential part, one from the Gaussian CDF (whose derivative is a Gaussian pdf). Lemma 6.1 says that the two *pdf* pieces are equal and cancel when summed with opposite signs. The surviving terms come only from the exponentials.

### 6.2 Derivative of $h_t$

Differentiate  $L_{\pm}$  using the product rule. Recall that  $\Phi'(x) = \phi(x)$ , so  $\frac{d}{dr} \Phi(r/\tau \pm a) = (1/\tau) \phi(r/\tau \pm a)$ .

$$L'_+(r) = -\frac{1}{\tilde{b}} e^{-r/\tilde{b}} \Phi\left(\frac{r}{\tau} - a\right) + \frac{1}{\tau} e^{-r/\tilde{b}} \phi\left(\frac{r}{\tau} - a\right), \quad (6.3)$$

$$L'_-(r) = +\frac{1}{\tilde{b}} e^{r/\tilde{b}} \Phi\left(-\frac{r}{\tau} - a\right) - \frac{1}{\tau} e^{r/\tilde{b}} \phi\left(\frac{r}{\tau} + a\right), \quad (6.4)$$

where we used  $\phi(-r/\tau - a) = \phi(r/\tau + a)$  by symmetry of  $\phi$  and a sign-flip under the inner derivative. Now sum:

$$L'_+(r) + L'_-(r) = -\frac{1}{\tilde{b}} L_+(r) + \frac{1}{\tilde{b}} L_-(r) + \frac{1}{\tau} \left[ e^{-r/\tilde{b}} \phi(r/\tau - a) - e^{r/\tilde{b}} \phi(r/\tau + a) \right]. \quad (6.5)$$

By Lemma 6.1, the bracket in the last term is exactly zero:

$$e^{-r/\tilde{b}} \phi\left(\frac{r}{\tau} - a\right) - e^{r/\tilde{b}} \phi\left(\frac{r}{\tau} + a\right) = e^{-a^2/2} \phi\left(\frac{r}{\tau}\right) - e^{-a^2/2} \phi\left(\frac{r}{\tau}\right) = 0. \quad (6.6)$$

Therefore

$$L'_+(r) + L'_-(r) = \frac{1}{\tilde{b}} [L_-(r) - L_+(r)], \quad (6.7)$$

and differentiating (5.17),

$$h'_t(r) = \frac{e^{a^2/2}}{2\tilde{b}^2} [L_-(r) - L_+(r)]. \quad (6.8)$$

### 6.3 The residual score

Dividing  $h'_t$  by  $h_t$  and cancelling the  $e^{a^2/2}/2$  factors:

$$\psi_t(r) := \frac{h'_t(r)}{h_t(r)} = \frac{1}{\tilde{b}} \frac{L_-(r) - L_+(r)}{L_+(r) + L_-(r)}. \quad (6.9)$$

**Theorem 6.2** (Joint score on the residual ray). *The joint score  $S(x, t) = \nabla_x \log p_t(x_0, x_1)$  of the  $K = 1$  model equals*

$$S(x, t) = \begin{pmatrix} -\frac{x_0 - \beta\mu_0}{v} - \rho\psi_t(r) \\ \psi_t(r) \end{pmatrix}, \quad r = x_1 - \nu - \rho x_0, \quad (6.10)$$

with  $\psi_t$  given by (6.9). In particular,  $\psi_t(r) \rightarrow \mp 1/\tilde{b}$  as  $r \rightarrow \pm\infty$ .

*Proof.* From the factorisation  $\log p_t(x_0, x_1) = \log g_t(x_0) + \log h_t(r)$ ,

$$\partial_{x_1} \log p_t = \partial_{x_1} \log h_t(r) = (\log h_t)'(r) \cdot \partial_{x_1} r = \psi_t(r), \quad (6.11)$$

$$\partial_{x_0} \log p_t = (\log g_t)'(x_0) + (\log h_t)'(r) \cdot \partial_{x_0} r = -\frac{x_0 - \beta\mu_0}{v} + \psi_t(r) \cdot (-\rho), \quad (6.12)$$

where we used  $\partial_{x_1} r = 1$  and  $\partial_{x_0} r = -\rho$ . Stacking gives (6.10).  $\square$

#### Tail behaviour of $\psi_t$

We prove the saturation claim rigorously. Focus on  $r \rightarrow +\infty$  (the other side is identical by symmetry of  $h_t$ ). Write the ratio in (6.9) as

$$\frac{L_-(r) - L_+(r)}{L_+(r) + L_-(r)} = \frac{L_-(r)/L_+(r) - 1}{L_-(r)/L_+(r) + 1} = \tanh\left(\frac{1}{2} \log(L_-(r)/L_+(r))\right), \quad (6.13)$$

where the second equality uses  $(x - 1)/(x + 1) = \tanh((\log x)/2)$  for  $x > 0$ . We have  $L_-(r)/L_+(r) = e^{2r/\tilde{b}} \Phi(-r/\tau - a)/\Phi(r/\tau - a)$ . As  $r \rightarrow +\infty$ :  $\Phi(r/\tau - a) \rightarrow 1$  and  $\Phi(-r/\tau - a) \rightarrow 0$ , but the factor  $e^{2r/\tilde{b}}$  grows so fast that we need to be careful. Using Mills' ratio  $\Phi(-x) \sim \phi(x)/x$  for  $x \rightarrow +\infty$ ,

$$\Phi(-r/\tau - a) \sim \frac{\phi(r/\tau + a)}{r/\tau + a} \rightarrow 0 \text{ super-exponentially in } r, \quad (6.14)$$

so  $\log L_-(r) \sim r/\tilde{b} - (r/\tau + a)^2/2 \sim -r^2/(2\tau^2)$  for large  $r$ , while  $\log L_+(r) \sim -r/\tilde{b}$ . Therefore

$$\log(L_-(r)/L_+(r)) \sim -\frac{r^2}{2\tau^2} + \frac{r}{\tilde{b}} \rightarrow -\infty \text{ as } r \rightarrow \infty. \quad (6.15)$$

Plugging into (6.13),  $\tanh((-\infty)/2) = -1$ , so  $\psi_t(r) \cdot \tilde{b} \rightarrow -1$ , i.e.  $\psi_t(r) \rightarrow -1/\tilde{b}$ . By  $r \mapsto -r$  symmetry,  $\psi_t(r) \rightarrow +1/\tilde{b}$  as  $r \rightarrow -\infty$ .  $\square$

*Interpretation* (Why the score saturates). For a Gaussian, the tails are so thin that escaping them costs quadratic energy, and the restoring force – the score – grows linearly. For a Laplace, the tails are only exponential; the cost of escape grows linearly, and the restoring force is bounded. Here, even with a Gaussian contribution  $G_t$  added to  $\beta\varepsilon$ , the Laplace tail dominates at large  $|r|$  and the score saturates at  $\pm 1/\tilde{b} = \pm 1/(\beta b)$ . Saturation is the signature of sub-Gaussian tails; any diffusion model that tries to represent a Laplace-like innovation with a Gaussian score will be *locally* wrong in the tails by a factor growing in  $|r|$ .

## Chapter 7

# Curvature theorem

The residual score  $\psi_t(r)$  depends only on  $r$ . The Hessian  $H_t(x)$  is obtained by differentiating the score and thus inherits this dependence: all of its entries are proportional to a single scalar function  $\kappa_t(r) := -\psi'_t(r)$ . We show this explicitly and then note the strong low-rank consequence.

### 7.1 Componentwise computation

From Theorem 6.2, we have  $S_0(x) = -(x_0 - \beta\mu_0)/v - \rho\psi_t(r)$  and  $S_1(x) = \psi_t(r)$ . The Hessian is

$$H_t(x) = -\nabla_x^2 \log p_t = - \begin{pmatrix} \partial_{x_0} S_0 & \partial_{x_1} S_0 \\ \partial_{x_0} S_1 & \partial_{x_1} S_1 \end{pmatrix}. \quad (7.1)$$

Using  $\partial_{x_0} r = -\rho$  and  $\partial_{x_1} r = 1$ , and writing  $\psi'_t(r) = -\kappa_t(r)$ ,

$$\partial_{x_1} S_1 = \psi'_t(r) = -\kappa_t(r), \quad (7.2)$$

$$\partial_{x_0} S_1 = \psi'_t(r) \cdot (-\rho) = \rho\kappa_t(r), \quad (7.3)$$

$$\partial_{x_1} S_0 = -\rho\psi'_t(r) = \rho\kappa_t(r), \quad (7.4)$$

$$\partial_{x_0} S_0 = -\frac{1}{v} - \rho\psi'_t(r) \cdot (-\rho) = -\frac{1}{v} - \rho^2\kappa_t(r) \quad (\text{wait, with the sign of } \psi'_t). \quad (7.5)$$

Let us redo the last one carefully:  $S_0 = -(x_0 - \beta\mu_0)/v - \rho\psi_t(r)$ , so  $\partial_{x_0} S_0 = -1/v - \rho\psi'_t(r) \cdot \partial_{x_0} r = -1/v - \rho\psi'_t(r) \cdot (-\rho) = -1/v + \rho^2\psi'_t(r) = -1/v - \rho^2\kappa_t(r)$ . Hence

$$H_t(x) = - \begin{pmatrix} -1/v - \rho^2\kappa_t(r) & \rho\kappa_t(r) \\ \rho\kappa_t(r) & -\kappa_t(r) \end{pmatrix} = \begin{pmatrix} 1/v + \rho^2\kappa_t(r) & -\rho\kappa_t(r) \\ -\rho\kappa_t(r) & \kappa_t(r) \end{pmatrix}. \quad (7.6)$$

**Theorem 7.1** (Rank-1 structure of the curvature,  $K=1$ ).

$$H_t(x) = \underbrace{\begin{pmatrix} 1/v & 0 \\ 0 & 0 \end{pmatrix}}_{=: Q_0} + \kappa_t(r) M_\rho, \quad M_\rho := \begin{pmatrix} \rho^2 & -\rho \\ -\rho & 1 \end{pmatrix}. \quad (7.7)$$

The matrix  $M_\rho$  has rank one, with  $M_\rho = nn^\top$  where  $n = (\rho, -1)^\top$ . Its eigenvalues are  $\{0, 1 + \rho^2\}$ , with eigenvectors  $\propto (1, \rho)^\top$  (kernel) and  $n$  respectively. Consequently, for fixed  $t$ , the curvature field  $H_t(x)$  traces a one-parameter affine ray in the space of  $2 \times 2$  symmetric matrices, parameterised by the scalar  $\kappa_t(r)$ .

*Proof.* Equation (7.7) is just a rearrangement of (7.6). For  $M_\rho$ :  $\det M_\rho = \rho^2 - \rho^2 = 0$ , so  $\text{rank} \leq 1$ , and the trace is  $1 + \rho^2 > 0$ , so rank is exactly 1. The non-zero eigenvalue is the trace  $1 + \rho^2$ ; the corresponding eigenvector is  $n = (\rho, -1)^\top$  (check:  $M_\rho n = (\rho^3 + \rho, -\rho^2 - 1)^\top = (1 + \rho^2)(\rho, -1)^\top$ ).  $\square$

*Interpretation.* The Gaussian case has  $H_t = \Sigma_t^{-1}$  constant in  $x$ : the curvature field is a single matrix. The Laplace case has  $H_t$  varying with  $x$ , but only along the 1D ray indexed by the scalar  $\kappa_t(r)$ . This is the non-Gaussian generalisation of “precision matrix”: the field is not a single matrix, but a smooth curve of matrices with rank-1 update, governed by the scalar function  $r \mapsto \kappa_t(r)$ .

## 7.2 Alternative expression for $\kappa_t$ : the ODE route

The formula  $\kappa_t(r) = -\psi_t'(r)$  can be turned into an explicit expression that avoids differentiating the tanh-type formula (6.9). The trick is the identity

$$\kappa_t(r) = -\frac{d^2}{dr^2} \log h_t(r) = -\frac{h_t''(r)}{h_t(r)} + \left(\frac{h_t'(r)}{h_t(r)}\right)^2 = \psi_t(r)^2 - \frac{h_t''(r)}{h_t(r)}. \quad (7.8)$$

We will show in Chapter 8 that  $h_t$  satisfies the linear ODE

$$(1 - \tilde{b}^2 \partial_r^2) h_t(r) = \phi_\tau(r), \quad \phi_\tau(r) := \frac{1}{\sqrt{2\pi\tau^2}} \exp\left(-\frac{r^2}{2\tau^2}\right). \quad (7.9)$$

Solving for  $h_t''$ ,

$$h_t''(r) = \frac{h_t(r) - \phi_\tau(r)}{\tilde{b}^2}. \quad (7.10)$$

Plugging into (7.8),

$$\boxed{\kappa_t(r) = \psi_t(r)^2 - \frac{1}{\tilde{b}^2} + \frac{1}{\tilde{b}^2} \frac{\phi_\tau(r)}{h_t(r)}}. \quad (7.11)$$

This is the form we will use to prove the Mills-ratio closed form of  $\kappa_t(0)$  in Chapter 9. It is also the form most convenient for numerical evaluation, since the tanh-type formula for  $\psi_t$  is stable and (7.11) avoids differentiating it.

*Remark* (Sanity check at the tails). As  $r \rightarrow \pm\infty$ :  $\psi_t(r) \rightarrow \mp 1/\tilde{b}$ , so  $\psi_t^2 \rightarrow 1/\tilde{b}^2$ , cancelling the  $-1/\tilde{b}^2$ . Meanwhile  $\phi_\tau(r)/h_t(r) \rightarrow 0$  because  $h_t$  has exponential Laplace tails whereas  $\phi_\tau$  has Gaussian tails that decay strictly faster. Therefore  $\kappa_t(\pm\infty) = 0$ , consistent with the saturation of  $\psi_t$ .



## Part III

# Fourier representation

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The results of Part II were obtained in real space: convolution in Section 5, chain rule in Chapter 6, linear algebra in Chapter 7. The Fourier picture tells the same story but in a cleaner geometric language: the characteristic function of  $p_t$  factorises explicitly, the non-Gaussian content collapses onto a single innovation-frequency axis, and the residual density satisfies a genuinely linear ODE. We now derive all of this from scratch.

## Chapter 8

# Characteristic function of the $K=1$ law

### 8.1 Clean characteristic function

The clean joint law of  $(A_0, A_1)$  has characteristic function

$$\hat{p}_0(q_0, q_1) := \mathbb{E}\left[e^{i(q_0 A_0 + q_1 A_1)}\right] = \mathbb{E}\left[e^{iq_0 A_0 + iq_1(\alpha A_0 + \varepsilon)}\right] = \mathbb{E}\left[e^{i(q_0 + \alpha q_1)A_0}\right] \mathbb{E}\left[e^{iq_1 \varepsilon}\right], \quad (8.1)$$

where we used independence of  $A_0$  and  $\varepsilon$ . The two expectations are standard:

$$\mathbb{E}\left[e^{i(q_0 + \alpha q_1)A_0}\right] = \exp\left(i\mu_0(q_0 + \alpha q_1) - \frac{\sigma_0^2}{2}(q_0 + \alpha q_1)^2\right), \quad (8.2)$$

$$\mathbb{E}\left[e^{iq_1 \varepsilon}\right] = \frac{1}{1 + b^2 q_1^2}, \quad \varepsilon \sim \text{Lap}(0, b). \quad (8.3)$$

**Why this step?** The characteristic function of  $\text{Lap}(0, b)$  is

$$\int_{\mathbb{R}} \frac{1}{2b} e^{-|u|/b} e^{iqu} du = \frac{1}{2b} \left[ \int_0^\infty e^{(iq-1/b)u} du + \int_{-\infty}^0 e^{(iq+1/b)u} du \right] = \frac{1}{1 + b^2 q^2}.$$

The rational form  $1/(1 + b^2 q^2)$  is what makes Laplace such a tractable non-Gaussian noise: it is the simplest nontrivial example of an “exponential of a rational pole” structure in Fourier space.

Collecting,

$$\hat{p}_0(q_0, q_1) = \frac{1}{1 + b^2 q_1^2} \exp\left(i\mu_0(q_0 + \alpha q_1) - \frac{\sigma_0^2}{2}(q_0 + \alpha q_1)^2\right). \quad (8.4)$$

### 8.2 OU corruption in Fourier space

The channel  $X_k = \beta A_k + \sqrt{\Delta} Z_k$  with  $Z_k \perp A$  implies, for the joint characteristic function  $\hat{p}_t(k_0, k_1) := \mathbb{E}[e^{i(k_0 X_0 + k_1 X_1)}]$ :

$$\begin{aligned} \hat{p}_t(k_0, k_1) &= \mathbb{E}\left[e^{i(k_0(\beta A_0 + \sqrt{\Delta} Z_0) + k_1(\beta A_1 + \sqrt{\Delta} Z_1))}\right] \\ &= \mathbb{E}\left[e^{i\beta(k_0 A_0 + k_1 A_1)}\right] \cdot \mathbb{E}\left[e^{i\sqrt{\Delta}(k_0 Z_0 + k_1 Z_1)}\right] \\ &= \hat{p}_0(\beta k_0, \beta k_1) \cdot e^{-\Delta(k_0^2 + k_1^2)/2}, \end{aligned} \quad (8.5)$$

using independence of  $A$  from  $(Z_0, Z_1)$  and the standard Gaussian CF  $\mathbb{E}[e^{iqZ}] = e^{-q^2/2}$ . Substituting (8.4) with  $(q_0, q_1) = (\beta k_0, \beta k_1)$  and writing  $\tilde{b} = \beta b$ ,

$$\hat{p}_t(k_0, k_1) = \frac{1}{1 + \tilde{b}^2 k_1^2} \exp\left(i\beta\mu_0(k_0 + \alpha k_1) - \frac{\beta^2\sigma_0^2}{2}(k_0 + \alpha k_1)^2 - \frac{\Delta}{2}(k_0^2 + k_1^2)\right). \quad (8.6)$$

**Theorem 8.1** (Fourier representation,  $K=1$ ). *The characteristic function of the noisy joint law is given by (8.6). In the rotated innovation-frequency coordinates*

$$\ell_0 := k_0 + \alpha k_1, \quad \ell_1 := k_1, \quad (8.7)$$

(equivalently,  $k_0 = \ell_0 - \alpha \ell_1$ ,  $k_1 = \ell_1$ ), the characteristic function has the form

$$\hat{q}_t(\ell_0, \ell_1) := \hat{p}_t(\ell_0 - \alpha \ell_1, \ell_1) = \underbrace{e^{i\beta\mu_0\ell_0 - \frac{\beta^2\sigma_0^2}{2}\ell_0^2 - \frac{\Delta}{2}[(\ell_0 - \alpha\ell_1)^2 + \ell_1^2]}}_{\text{Gaussian sector}} \cdot \underbrace{\frac{1}{1 + \tilde{b}^2\ell_1^2}}_{\text{rational sector}}. \quad (8.8)$$

The non-Gaussian content is a function of  $\ell_1$  alone; the Gaussian sector is a nondegenerate quadratic form in  $(\ell_0, \ell_1)$ .

*Proof.* Equation (8.6) is derived above. Substitute  $k_0 + \alpha k_1 = \ell_0$  and  $k_1 = \ell_1$ ; also  $k_0^2 + k_1^2 = (\ell_0 - \alpha \ell_1)^2 + \ell_1^2$ . The rational factor becomes  $1/(1 + \tilde{b}^2\ell_1^2)$ . No further simplification is needed.  $\square$

*Interpretation* (Why this rotation is natural). The clean chain has the causal decomposition  $A_0, \varepsilon = A_1 - \alpha A_0$  with  $A_0, \varepsilon$  independent. Their Fourier duals are  $k_0 + \alpha k_1$  (the conjugate of  $A_0$ ) and  $k_1$  (the conjugate of  $\varepsilon$ ). The rotation (8.7) is *exactly* this causal coordinate change, rewritten in Fourier space. After the rotation, the Laplace innovation contributes one rational factor in  $\ell_1$  only, decoupled from the site frequency  $\ell_0$ .

### 8.3 From characteristic function to residual ODE

We now derive (7.9), the linear ODE satisfied by the residual density  $h_t(r)$ . Start from the observation that  $R_t = \beta\varepsilon + G_t$  has characteristic function

$$\hat{h}_t(\ell) := \mathbb{E}[e^{i\ell R_t}] = \mathbb{E}[e^{i\ell\beta\varepsilon}] \mathbb{E}[e^{i\ell G_t}] = \frac{1}{1 + \tilde{b}^2\ell^2} \cdot e^{-\tau^2\ell^2/2}. \quad (8.9)$$

Multiply both sides by  $(1 + \tilde{b}^2\ell^2)$ :

$$(1 + \tilde{b}^2\ell^2) \hat{h}_t(\ell) = e^{-\tau^2\ell^2/2}. \quad (8.10)$$

Invert the Fourier transform. Under the convention  $h_t(r) = \frac{1}{2\pi} \int e^{-i\ell r} \hat{h}_t(\ell) d\ell$ , multiplication by  $i\ell$  corresponds to  $\partial_r$ ; in particular  $(i\ell)^2 = -\ell^2$  corresponds to  $\partial_r^2$ , so multiplication by  $\ell^2$  corresponds to  $-\partial_r^2$ . Applying the inverse transform to both sides of (8.10):

$$(1 - \tilde{b}^2\partial_r^2) h_t(r) = \phi_\tau(r), \quad \phi_\tau(r) := \frac{1}{\sqrt{2\pi\tau^2}} e^{-r^2/(2\tau^2)}. \quad (8.11)$$

This is (7.9), rigorously derived.

*Remark* (Check: the clean limit). At  $t = 0$ :  $\tau = 0$ , so  $\phi_\tau(r)$  becomes a Dirac delta  $\delta(r)$ ; the ODE reduces to  $(1 - b^2\partial_r^2)h_0 = \delta$ , whose solution (with decay at infinity) is the Green's function  $h_0(r) = (1/2b)e^{-|r|/b}$ , the clean Laplace density. Good.

*Remark* (Check: the far- $t$  limit). As  $t \rightarrow \infty$ :  $\tilde{b} \rightarrow 0$ , so the ODE reduces to  $h_\infty(r) = \phi_\tau(r)$  (a pure Gaussian with variance  $\tau^2 \rightarrow 1$ ). This recovers the Gaussian benchmark in the fully-noised regime.

## 8.4 Recovering $h_t$ from the ODE: an alternative route

The ODE (8.11) gives an alternative way of arriving at the closed form (5.15), bypassing the direct convolution. The Green's function of  $1 - \tilde{b}^2 \partial_r^2$  that decays at infinity is  $(2\tilde{b})^{-1} e^{-|r|/\tilde{b}}$ , so

$$h_t(r) = \int_{\mathbb{R}} \frac{1}{2\tilde{b}} e^{-|r-u|/\tilde{b}} \phi_\tau(u) du. \quad (8.12)$$

This is not our original convolution, but a dual one: a Laplace density convolved against a Gaussian. By the symmetry of convolution, however, this is the same object, and working through the completion-of-squares in  $u$  (instead of in  $r$ ) gives (5.15) with the roles of  $L_+$  and  $L_-$  symmetrised.

**Why this step?** Why the ODE matters: from (8.11) we read off  $h_t'' = (h_t - \phi_\tau)/\tilde{b}^2$  without having to differentiate (5.15) twice. This is the single fact that made the alternative expression (7.11) for  $\kappa_t(r)$  possible – and that alternative expression is exactly what delivers the Mills-ratio closed form at  $r = 0$  in the next part.

## 8.5 A diagnostic: where nonlinearity enters

In Fourier space, everything is *linear*: the characteristic function, the rational rational factor, the ODE. The *nonlinearity* of the score and curvature comes from a single operation: taking the logarithm of the real-space density.

- $h_t$  is a linear combination of exponential-times-Gaussian-CDF terms.
- $\log h_t$  is not linear in those terms; it combines them via a log-sum-exp, which gives a tanh-type saturating score.
- $\partial_r \log h_t = h_t'/h_t$  is therefore bounded (saturates);  $\partial_r^2 \log h_t$  is concentrated near  $r = 0$  (where the ratio  $\phi_\tau/h_t$  peaks) and vanishes at infinity.

This is the cleanest identification of where non-Gaussianity lives in the problem. It foreshadows the broader story of the thesis: *the density is linearly structured in Fourier; the score is where nonlinearity takes hold, and it is concentrated in the innovation direction.*

## Part IV

# Scalar observables

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The structure theorems of Part II describe the entire residual density  $h_t(r)$ , the entire score  $\psi_t(r)$ , and the entire curvature field  $\kappa_t(r)$ . These are *functions* of  $r$ ; to get finite-dimensional diagnostics that we can track as  $t$  varies – and that play the role of the Gaussian precision-matrix mass – we extract scalar summaries. Each one is a functional of  $h_t$ . Our goal is to compute each in closed form and to interpret what it measures.

## Chapter 9

# Centre curvature: the Mills-ratio closed form

The most important scalar summary is the curvature at the origin,  $\kappa_t(0)$ . We state and prove in full the identity that expresses it through Mills' ratio, and then interpret its limits.

### 9.1 Statement

**Theorem 9.1** (Closed form for  $\kappa_t(0)$ ). *Let  $\phi, \Phi$  denote the standard normal pdf and cdf, and let*

$$R(a) := \frac{\Phi(-a)}{\phi(a)}, \quad a \in \mathbb{R}, \quad (9.1)$$

*be Mills' ratio of the standard normal. Then, with  $a = \tau/\tilde{b}$ ,*

$$\boxed{\kappa_t(0) \tilde{b}^2 = \frac{1}{a R(a)} - 1.} \quad (9.2)$$

*Equivalently,*

$$\kappa_t(0) = \frac{1}{\tilde{b}^2} \left( \frac{1}{a R(a)} - 1 \right). \quad (9.3)$$

**Why this step?** Why this is the right object:  $\kappa_t(0) \tilde{b}^2$  is dimensionless (both sides have dimension of length squared, canceling). It depends on  $t$  only through the single scalar  $a(t) = \tau(t)/\tilde{b}(t)$ , which monotonically increases from 0 at  $t = 0$  to  $+\infty$  as  $t \rightarrow \infty$ . Thus the entire one-parameter family of residual densities is compressed into a single dimensionless curve.

### 9.2 Proof

The proof has three elementary ingredients: the alternative formula for  $\kappa_t$  from Section 7, the closed form for  $h_t(0)$ , and Mills' ratio at a single point.

**Step 1:**  $\psi_t(0) = 0$ . By direct inspection of (5.15),  $h_t(-r) = h_t(r)$ : swap the two terms using  $\Phi(r/\tau - a) \leftrightarrow \Phi(-r/\tau - a)$  under  $r \mapsto -r$ . So  $h_t$  is an even function, and its derivative at 0 is zero:

$$\psi_t(0) = \frac{h'_t(0)}{h_t(0)} = 0. \quad (9.4)$$

**Step 2: the alternative formula evaluated at 0.** From (7.11),

$$\kappa_t(0) = \psi_t(0)^2 - \frac{1}{\tilde{b}^2} + \frac{1}{\tilde{b}^2} \frac{\phi_\tau(0)}{h_t(0)} = -\frac{1}{\tilde{b}^2} + \frac{1}{\tilde{b}^2} \frac{\phi_\tau(0)}{h_t(0)}. \quad (9.5)$$

It remains to evaluate the ratio  $\phi_\tau(0)/h_t(0)$ .



**Step 3: compute  $h_t(0)$ .** From (5.15) at  $r = 0$ :

$$h_t(0) = \frac{e^{a^2/2}}{2\tilde{b}} [L_+(0) + L_-(0)] = \frac{e^{a^2/2}}{2\tilde{b}} [e^0\Phi(-a) + e^0\Phi(-a)] = \frac{e^{a^2/2}}{\tilde{b}}\Phi(-a). \quad (9.6)$$

(We used  $L_+(0) = \Phi(-a)$  and  $L_-(0) = \Phi(-a)$  from (5.16).)

**Step 4: compute  $\phi_\tau(0)/h_t(0)$ .**  $\phi_\tau(0) = 1/\sqrt{2\pi\tau^2}$ , and rewriting using  $\tau = a\tilde{b}$ :

$$\phi_\tau(0) = \frac{1}{\sqrt{2\pi} a\tilde{b}}. \quad (9.7)$$

Dividing,

$$\frac{\phi_\tau(0)}{h_t(0)} = \frac{1/(\sqrt{2\pi} a\tilde{b})}{e^{a^2/2}\Phi(-a)/\tilde{b}} = \frac{1}{a\sqrt{2\pi} e^{a^2/2}\Phi(-a)} = \frac{\phi(a)}{a\phi(a)\sqrt{2\pi} e^{a^2/2}\Phi(-a)}. \quad (9.8)$$

But  $\phi(a) = (2\pi)^{-1/2}e^{-a^2/2}$ , so  $(2\pi)^{1/2}e^{a^2/2} = 1/\phi(a)$ , and

$$\frac{\phi_\tau(0)}{h_t(0)} = \frac{\phi(a)}{a\Phi(-a)} = \frac{1}{aR(a)}, \quad (9.9)$$

where we used (9.1).

**Step 5: assemble.** Substituting (9.9) into (9.5),

$$\kappa_t(0)\tilde{b}^2 = -1 + \frac{1}{aR(a)} = \frac{1}{aR(a)} - 1, \quad (9.10)$$

which is (9.2).  $\square$

### 9.3 Sanity check: the two extreme limits

Mills' ratio has well-known expansions at  $a \rightarrow 0$  and  $a \rightarrow \infty$ . Using them, we recover the clean Laplace and Gaussian limits.

**Clean limit  $t \rightarrow 0^+$  ( $a \rightarrow 0^+$ ).** At  $a = 0$ :  $\phi(0) = 1/\sqrt{2\pi}$ ,  $\Phi(0) = 1/2$ , so  $R(0) = \sqrt{2\pi}/2 = \sqrt{\pi}/2$ . Expanding,

$$R(a) = \sqrt{\pi}/2 (1 - a\sqrt{2/\pi} + O(a^2)), \quad a \rightarrow 0. \quad (9.11)$$

Hence

$$\frac{1}{aR(a)} = \frac{1}{a\sqrt{\pi}/2} (1 + a\sqrt{2/\pi} + O(a^2)) = \frac{\sqrt{2/\pi}}{a} + 1 + O(a). \quad (9.12)$$

Therefore

$$\kappa_t(0)\tilde{b}^2 \sim \frac{\sqrt{2/\pi}}{a} \rightarrow +\infty \quad (a \rightarrow 0^+), \quad (9.13)$$

i.e. the centre curvature diverges as  $1/a = \tilde{b}/\tau$  when  $t \rightarrow 0$ . This is the  $r = 0$  Laplace cusp: the clean Laplace density has a corner at 0, so the log-density has infinite negative curvature there.

**Fully-noised limit**  $t \rightarrow \infty$  ( $a \rightarrow \infty$ ). For large  $a$ ,

$$R(a) \sim \frac{1}{a} - \frac{1}{a^3} + O(a^{-5}), \quad (9.14)$$

hence  $aR(a) \sim 1 - 1/a^2 + \dots$ , and  $1/(aR(a)) \sim 1 + 1/a^2$ . Subtracting 1:

$$\kappa_t(0) \tilde{b}^2 \sim \frac{1}{a^2} = \frac{\tilde{b}^2}{\tau^2}. \quad (9.15)$$

Thus  $\kappa_t(0) \sim 1/\tau^2$ . Compare to the Gaussian-benchmark curvature  $\kappa_t^{(G)} = 1/\tau_G^2 = 1/(\tau^2 + 2\tilde{b}^2)$ : for large  $a$ ,  $\tau^2 \gg \tilde{b}^2$ , and both agree to leading order.

*Interpretation* (What the Mills-ratio form buys us). Equation (9.2) is not just a compact formula: it is the explicit statement that the centre curvature is a universal function of a single dimensionless ratio  $a$ . As  $a$  sweeps from 0 to  $\infty$  along diffusion time,  $\kappa_t(0)\tilde{b}^2$  sweeps from  $+\infty$  to 0, passing through every positive value exactly once (monotonicity is not proved here but follows from the fact that  $aR(a)$  is strictly increasing). Every finite-time non-Gaussianity of the centre of the density is encoded in this single scalar curve, and the Gaussian benchmark simply reads as a different universal function of  $a$ , differing by a correction of order  $1/a^4$  at large  $a$ .

## Chapter 10

# Gaussian benchmark and the centre-benchmark gap

To calibrate non-Gaussianity, we compare the Laplace benchmark against a *matched-variance* Gaussian model. This gives a rigorous notion of “how far is the true curvature from the Gaussian fit of the same two-point statistics.”

### 10.1 The matched Gaussian innovation

Replace  $\varepsilon \sim \text{Lap}(0, b)$  by  $\varepsilon^{(G)} \sim \mathcal{N}(0, 2b^2)$ . Since  $\text{Var}(\varepsilon) = 2b^2 = \text{Var}(\varepsilon^{(G)})$ , the two innovations have identical second-order statistics. In the Laplace benchmark,  $R_t = \beta\varepsilon + G_t$  has variance  $2\tilde{b}^2 + \tau^2 =: \tau_G^2$ ; in the matched Gaussian benchmark, the corresponding residual  $R_t^{(G)}$  has the same variance but is Gaussian:

$$R_t^{(G)} \sim \mathcal{N}(0, \tau_G^2), \quad \tau_G^2 := 2\tilde{b}^2 + \tau^2. \quad (10.1)$$

The joint noisy law is jointly Gaussian and the curvature is constant in  $x$ :

$$H_t^{(G)} = \begin{pmatrix} 1/v + \rho^2/\tau_G^2 & -\rho/\tau_G^2 \\ -\rho/\tau_G^2 & 1/\tau_G^2 \end{pmatrix} = \begin{pmatrix} 1/v & 0 \\ 0 & 0 \end{pmatrix} + \kappa_t^{(G)} M_\rho, \quad \kappa_t^{(G)} := \frac{1}{\tau_G^2}. \quad (10.2)$$

**Why this step?** Matching  $\text{Var}(\varepsilon) = \text{Var}(\varepsilon^{(G)}) = 2b^2$  means the two benchmarks cannot be told apart by any two-point statistic of the residual. Everything that distinguishes them – the cusp at  $r = 0$ , the saturating score, the location-dependent curvature – is by construction higher-order.

### 10.2 The centre-benchmark gap

The residual curvature at  $r = 0$  of the Laplace model is  $\kappa_t(0) = (1/\tilde{b}^2)(1/(aR(a)) - 1)$ . The matched Gaussian gives constant  $\kappa_t^{(G)} = 1/(\tau^2 + 2\tilde{b}^2) = 1/(\tilde{b}^2(a^2 + 2))$ . Their gap is

$$\delta\kappa_t := \kappa_t(0) - \kappa_t^{(G)} = \frac{1}{\tilde{b}^2} \left[ \frac{1}{aR(a)} - 1 - \frac{1}{a^2 + 2} \right]. \quad (10.3)$$

**Small  $a$  (small  $t$ ): the cusp dominates.**  $1/(aR(a)) \sim \sqrt{2/\pi}/a$  and  $1/(a^2 + 2) \rightarrow 1/2$ , so  $\delta\kappa_t \sim (\tilde{b}^2)^{-1} \sqrt{2/\pi}/a$ , i.e. diverges like  $1/a$ . The Gaussian fit misses the Laplace cusp catastrophically in the clean regime.

**Large  $a$  (large  $t$ ): the gap closes.** Use  $1/(aR(a)) \sim 1 + 1/a^2 + 3/a^4 + \dots$  (from the expansion  $R(a) = 1/a - 1/a^3 + 3/a^5 - \dots$ ) and  $1/(a^2 + 2) = 1/a^2 - 2/a^4 + O(a^{-6})$ :

$$\begin{aligned} \frac{1}{aR(a)} - 1 - \frac{1}{a^2 + 2} &= \left( \frac{1}{a^2} + \frac{3}{a^4} + O(a^{-6}) \right) - \left( \frac{1}{a^2} - \frac{2}{a^4} + O(a^{-6}) \right) \\ &= \frac{5}{a^4} + O(a^{-6}). \end{aligned} \tag{10.4}$$

So  $\delta\kappa_t \sim 5/(\tilde{b}^2 a^4) = 5\tilde{b}^2/\tau^4$ , which goes to zero as  $t \rightarrow \infty$ . The gap closes quartically.

*Interpretation* (Where the Gaussian benchmark is honest). For small  $t$  (strong non-Gaussianity), the Gaussian benchmark *underestimates* centre curvature by a factor growing like  $\tilde{b}/\tau$ ; a score trained against a Gaussian fit will miss the cusp entirely. For large  $t$ , the gap closes extremely fast ( $\tilde{b}^2/\tau^4$ ), so non-Gaussian fine-tuning yields negligible benefit in the late-diffusion regime. This is the quantitative underpinning of the folklore that diffusion models “don’t need to care about non-Gaussianity at late times.”

# Chapter 11

## Kurtosis and bandwidth

### 11.1 The fourth cumulant of $R_t$

**Proposition 11.1** (Residual fourth cumulant). *Let  $\kappa_4(Y)$  denote the fourth cumulant of a real random variable  $Y$ . Then*

$$\kappa_4(R_t) = 12\beta^4 b^4, \quad (11.1)$$

and consequently the excess kurtosis of  $R_t$  is

$$\frac{\kappa_4(R_t)}{\text{Var}(R_t)^2} = \frac{12\beta^4 b^4}{(2\tilde{b}^2 + \tau^2)^2} = \frac{12\tilde{b}^4}{\tilde{b}^4(a^2 + 2)^2} = \frac{12}{(a^2 + 2)^2} = \frac{3}{(1 + a^2/2)^2}. \quad (11.2)$$

*Proof.*  $R_t = \beta\varepsilon + G_t$  is a sum of independent variables, so cumulants add:  $\kappa_4(R_t) = \kappa_4(\beta\varepsilon) + \kappa_4(G_t)$ . The Gaussian  $G_t$  has  $\kappa_4 = 0$ . The fourth cumulant of  $\beta\varepsilon$  scales as  $\beta^4$  times that of  $\varepsilon \sim \text{Lap}(0, b)$ . For  $\text{Laplace}(0, b)$ , the fourth moment is  $24b^4$  and the second moment is  $2b^2$ , so  $\kappa_4 = m_4 - 3m_2^2 = 24b^4 - 3(2b^2)^2 = 12b^4$ . (Here  $\kappa_3 = 0$  by symmetry.) Hence  $\kappa_4(R_t) = 12\beta^4 b^4$ . Substituting  $\text{Var}(R_t) = 2\tilde{b}^2 + \tau^2$  and  $\tilde{b} = \beta b$  gives (11.2).  $\square$

*Interpretation* (Non-Gaussianity persistence). At  $t = 0$  ( $a = 0$ ): excess kurtosis is 3, the Laplace value. At  $t \rightarrow \infty$  ( $a \rightarrow \infty$ ): excess kurtosis  $\rightarrow 0$ , the Gaussian limit. The decay is  $\sim 12/a^4$ , the same quartic rate as the centre-benchmark gap. This is not a coincidence: both quantities are measuring the same thing – the fourth-order departure from Gaussianity of  $R_t$  – just through different functionals (one pointwise at  $r = 0$ , one integrated).

### 11.2 Full width at half maximum of $\kappa_t$

Since  $\kappa_t(r)$  is an even function peaked at  $r = 0$  and decaying to zero at  $\pm\infty$ , the characteristic width of the curvature band is

$$\text{FWHM}(t) := 2r^*(t), \quad \kappa_t(r^*) = \frac{1}{2}\kappa_t(0). \quad (11.3)$$

This is the non-Gaussian generalisation of the “tridiagonal mass” bandwidth  $b_{0.95}(t)$  used in the Gaussian case: it quantifies how localised the curvature is to the residual-ray centre.

**Why this step?** Why FWHM rather than another width: the function  $\kappa_t(r)$  has heavy tails (they decay to zero) but a clear peak at 0. Second moment is dominated by the tails and does not measure concentration; variance of the corresponding curvature mass is ill-defined; FWHM is well-defined, robust, and easy to compute from the closed form.

**Two natural length scales.** The residual density  $h_t(r)$  is built from two elementary scales:  $\tilde{b}$  (Laplace scale of the rescaled innovation) and  $\tau$  (Gaussian scale of the smoothing). Numerical experiments (Figure 4 of the main memo) confirm:

- $\text{FWHM}(t) \approx \tilde{b}$  for small  $t$  (Laplace regime,  $a \ll 1$ );
- $\text{FWHM}(t) \approx \tau$  for large  $t$  (Gaussian regime,  $a \gg 1$ );
- crossover around  $a = \tau/\tilde{b} \approx 1$ .

A closed form for  $\text{FWHM}(t)$  is possible in principle via the alternative formula (7.11) but is not illuminating; the scales  $\tilde{b}$  and  $\tau$  provide the right intuition.

## Part V

### The $K=2$ case

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The  $K = 1$  case is a pedagogical exercise: one innovation, one residual, one non-Gaussian scalar. The  $K = 2$  case is where the central question of the thesis starts to bite. Two innovations are coupled through the shared latent state  $A_1$ ; can we still factorise the exact density into a product of one-dimensional non-Gaussian bonds? The answer, as we will show in full, is *no*: there is a specific Gaussian cross-covariance,  $-\alpha\Delta$ , that forbids this factorisation at any finite  $t$ . But the *Markov locality pattern* survives: non-Gaussianity remains diagonal in innovation coordinates, and the Gaussian sector only couples nearest neighbours. We work out every step.



# Chapter 12

## Characteristic function for $K=2$

### A note on rotation conventions

Before diving in, we flag a convention change relative to Part III. Two different rotations of the frequency variables are natural in this problem, and we use each where it is cleanest.

**Causal (prior/innovation) rotation.** This is the Fourier dual of the real-space innovation coordinates  $(x_0, y_1, y_2, \dots)$  with  $y_k = x_k - \alpha x_{k-1}$ . It diagonalises the linear phase of the characteristic function (making it depend only on the first coordinate), but it leaves the Laplace rational factors on non-axis-aligned directions.

**Minimal (innovation) rotation.** This is the rotation that aligns the Laplace rational factors with the coordinate axes, but leaves the linear phase in a general direction.

At  $K = 1$  these two rotations coincide (there is only one innovation), and Part III uses the causal convention  $\ell_0 = k_0 + \alpha k_1$ ,  $\ell_1 = k_1$ . This choice highlights the one-dimensional non-Gaussian axis  $\ell_1$ .

At  $K = 2$  the two rotations differ. We will use the *minimal* convention  $\ell_0 = k_0$ ,  $\ell_1 = k_1 + \alpha k_2$ ,  $\ell_2 = k_2$  because it is the one that separates the two Laplace innovations onto distinct axes — and separating the innovations is precisely the content we need to test when asking whether the  $K = 1$  factorisation extends. A reader who prefers the fully-causal rotation will find it equivalent (the two differ by a further Gaussian shear in the site axis), and will obtain a slightly different  $M$  matrix whose off-diagonal structure is tridiagonal rather than having one block-column pattern; the cross-term  $M_{12} = -\alpha\Delta$  is the same in both conventions and is what matters for the obstruction theorem.

### 12.1 Clean characteristic function

The clean chain for  $K = 2$  has

$$A_0 \sim \mathcal{N}(\mu_0, \sigma_0^2), \quad A_1 = \alpha A_0 + \varepsilon_1, \quad A_2 = \alpha A_1 + \varepsilon_2, \quad (12.1)$$

with  $\varepsilon_1, \varepsilon_2 \stackrel{\text{iid}}{\sim} \text{Lap}(0, b)$  independent of  $A_0$ .

The clean characteristic function is

$$\hat{p}_0(q_0, q_1, q_2) = \mathbb{E} \left[ e^{i(q_0 A_0 + q_1 A_1 + q_2 A_2)} \right]. \quad (12.2)$$

Substitute  $A_1 = \alpha A_0 + \varepsilon_1$  and  $A_2 = \alpha^2 A_0 + \alpha \varepsilon_1 + \varepsilon_2$ :

$$q_0 A_0 + q_1 A_1 + q_2 A_2 = (q_0 + \alpha q_1 + \alpha^2 q_2) A_0 + (q_1 + \alpha q_2) \varepsilon_1 + q_2 \varepsilon_2. \quad (12.3)$$

By independence of  $A_0, \varepsilon_1, \varepsilon_2$ , the characteristic function factorises:

$$\hat{p}_0(q_0, q_1, q_2) = \underbrace{\mathbb{E}[e^{iqA_0}]}_{q=q_0+\alpha q_1+\alpha^2 q_2} \underbrace{\mathbb{E}[e^{i(q_1+\alpha q_2)\varepsilon_1}]}_{=1/(1+b^2(q_1+\alpha q_2)^2)} \underbrace{\mathbb{E}[e^{iq_2\varepsilon_2}]}_{=1/(1+b^2 q_2^2)}. \quad (12.4)$$

With the Gaussian CF  $\mathbb{E}[e^{iqA_0}] = \exp(i\mu_0 q - \sigma_0^2 q^2/2)$ ,

$$\hat{p}_0(q_0, q_1, q_2) = \frac{e^{i\mu_0 q - \sigma_0^2 q^2/2}}{(1 + b^2(q_1 + \alpha q_2)^2)(1 + b^2 q_2^2)}, \quad q := q_0 + \alpha q_1 + \alpha^2 q_2. \quad (12.5)$$

**Why this step?** Notice that the rational factor has two separate denominators: one for each innovation. The first,  $1 + b^2(q_1 + \alpha q_2)^2$ , comes from  $\varepsilon_1 = A_1 - \alpha A_0$ ; the second,  $1 + b^2 q_2^2$ , from  $\varepsilon_2 = A_2 - \alpha A_1$ . The *rotation* that separates them is not  $(q_0, q_1, q_2) \rightarrow (q_0, q_1, q_2)$  but a linear change of coordinates we will identify below.

## 12.2 OU corruption

The argument of Section 8.3 extends verbatim: the joint CF of the noisy chain is

$$\hat{p}_t(k_0, k_1, k_2) = \hat{p}_0(\beta k_0, \beta k_1, \beta k_2) e^{-\Delta(k_0^2 + k_1^2 + k_2^2)/2}. \quad (12.6)$$

Substituting (12.5) and writing  $\tilde{b} = \beta b$ ,  $\tilde{q} := \beta q = \beta(k_0 + \alpha k_1 + \alpha^2 k_2)$ :

$$\hat{p}_t(k_0, k_1, k_2) = \frac{\exp\left[i\beta\mu_0(k_0 + \alpha k_1 + \alpha^2 k_2) - \frac{\beta^2\sigma_0^2}{2}(k_0 + \alpha k_1 + \alpha^2 k_2)^2 - \frac{\Delta}{2}(k_0^2 + k_1^2 + k_2^2)\right]}{(1 + \tilde{b}^2(k_1 + \alpha k_2)^2)(1 + \tilde{b}^2 k_2^2)}. \quad (12.7)$$

## 12.3 Rotation to innovation-frequency coordinates

Define

$$\ell_0 := k_0, \quad \ell_1 := k_1 + \alpha k_2, \quad \ell_2 := k_2. \quad (12.8)$$

This change of variables is invertible with

$$k_0 = \ell_0, \quad k_1 = \ell_1 - \alpha \ell_2, \quad k_2 = \ell_2. \quad (12.9)$$

In the new coordinates:

- The innovation-2 rational factor is  $1/(1 + \tilde{b}^2 \ell_2^2)$  (unchanged, since  $k_2 = \ell_2$ ).
- The innovation-1 rational factor becomes  $1/(1 + \tilde{b}^2 \ell_1^2)$  (exactly, using  $k_1 + \alpha k_2 = \ell_1 - \alpha \ell_2 + \alpha \ell_2 = \ell_1$ ).
- The linear phase term  $k_0 + \alpha k_1 + \alpha^2 k_2$  becomes  $\ell_0 + \alpha(\ell_1 - \alpha \ell_2) + \alpha^2 \ell_2 = \ell_0 + \alpha \ell_1$ . *The  $\ell_2$  contribution cancels.*

The rational part is now *fully diagonal*: each Laplace factor depends on one innovation frequency only. The Gaussian sector is a quadratic form in  $(\ell_0, \ell_1, \ell_2)$ , which we compute carefully.

**Computing the Gaussian quadratic form.** Let  $\mathcal{Q}$  denote the quadratic form appearing in the exponent:

$$\mathcal{Q}(k_0, k_1, k_2) = \beta^2 \sigma_0^2 (k_0 + \alpha k_1 + \alpha^2 k_2)^2 + \Delta (k_0^2 + k_1^2 + k_2^2). \quad (12.10)$$

Under the rotation:

$$(k_0 + \alpha k_1 + \alpha^2 k_2)^2 = (\ell_0 + \alpha \ell_1)^2 = \ell_0^2 + 2\alpha \ell_0 \ell_1 + \alpha^2 \ell_1^2, \quad (12.11)$$

$$k_0^2 = \ell_0^2, \quad (12.12)$$

$$k_1^2 = (\ell_1 - \alpha \ell_2)^2 = \ell_1^2 - 2\alpha \ell_1 \ell_2 + \alpha^2 \ell_2^2, \quad (12.13)$$

$$k_2^2 = \ell_2^2. \quad (12.14)$$

Summing,

$$\begin{aligned} \mathcal{Q}(\ell_0, \ell_1, \ell_2) &= \beta^2 \sigma_0^2 (\ell_0^2 + 2\alpha \ell_0 \ell_1 + \alpha^2 \ell_1^2) \\ &\quad + \Delta (\ell_0^2 + \ell_1^2 - 2\alpha \ell_1 \ell_2 + \alpha^2 \ell_2^2 + \ell_2^2) \\ &= \underbrace{(\Delta + \beta^2 \sigma_0^2) \ell_0^2}_{M_{00}} + \underbrace{(\Delta + \alpha^2 \beta^2 \sigma_0^2) \ell_1^2}_{M_{11}} + \underbrace{\Delta (1 + \alpha^2) \ell_2^2}_{M_{22}} \\ &\quad + 2 \underbrace{\alpha \beta^2 \sigma_0^2 \ell_0 \ell_1}_{M_{01}} + 2 \underbrace{(-\alpha \Delta) \ell_1 \ell_2}_{M_{12}} + 2 \cdot 0 \cdot \ell_0 \ell_2. \end{aligned} \quad (12.15)$$

Thus  $\mathcal{Q}(\ell) = \ell^\top M \ell$  with

$$M = \begin{pmatrix} \Delta + \beta^2 \sigma_0^2 & \alpha \beta^2 \sigma_0^2 & 0 \\ \alpha \beta^2 \sigma_0^2 & \Delta + \alpha^2 \beta^2 \sigma_0^2 & -\alpha \Delta \\ 0 & -\alpha \Delta & \Delta(1 + \alpha^2) \end{pmatrix}. \quad (12.16)$$

**Why this step?** Notice three things in  $M$ : (i)  $M_{02} = 0$ : the site frequency  $\ell_0$  does not couple to the last innovation frequency  $\ell_2$ . This is the Fourier-space signature of Markov locality: the chain has only nearest-neighbour coupling. (ii)  $M_{01} = \alpha \beta^2 \sigma_0^2 \neq 0$ : the site frequency *does* couple to the first innovation frequency. This is the prior's effect on the first step of the chain. (iii)  $M_{12} = -\alpha \Delta \neq 0$ : the two innovation frequencies *do* couple. This is the obstruction we now examine.

## 12.4 Theorem: the characteristic function factorisation

**Theorem 12.1** (Fourier representation,  $K=2$ ). *In the rotated coordinates (12.8),*

$$\hat{q}_t(\ell) := \hat{p}_t(\ell_0, \ell_1 - \alpha \ell_2, \ell_2) = e^{i\beta \mu_0(\ell_0 + \alpha \ell_1) - \frac{1}{2} \ell^\top M \ell} \cdot \frac{1}{1 + \tilde{b}^2 \ell_1^2} \cdot \frac{1}{1 + \tilde{b}^2 \ell_2^2}, \quad (12.17)$$

with  $M$  given by (12.16).

*Proof.* Direct substitution, using the algebra above.  $\square$

*Remark* (The clean-chain limit). At  $t = 0$ :  $\Delta = 0$ ,  $\beta = 1$ , so  $M = \sigma_0^2 \text{diag}(1, \alpha^2, 0) + \dots$  and in fact  $M_{22} = 0$ , i.e. the Gaussian sector becomes degenerate in the  $\ell_2$  direction. This matches the fact that at  $t = 0$  the chain is Laplace-innovation-only in  $A_2 - \alpha A_1 = \varepsilon_2$ , with no Gaussian smoothing, and the rational factor  $1/(1 + \tilde{b}^2 \ell_2^2)$  is a pure Laplace distribution on its own.

# Chapter 13

## The obstruction to one-dimensional closure

### 13.1 Statement of the obstruction

For  $K = 1$ , the residual density  $h_t(r)$  was a function of a single coordinate  $r = x_1 - \nu - \rho x_0$ . Theorem 12.1 isolates two innovation frequencies  $\ell_1, \ell_2$ , whose *rational* factors are already separated. If we could further decouple their *Gaussian* contributions, we would get a full product factorisation into a site mode (in  $\ell_0$ ) and two independent bonds (in  $\ell_1$  and  $\ell_2$ ). The question is whether this is possible.

**Theorem 13.1** ( $K=2$  obstruction). *For any  $\alpha \neq 0$  and  $t > 0$ , the joint noisy law  $p_t(x_0, x_1, x_2)$  does not admit a factorisation of the form*

$$p_t(x_0, x_1, x_2) \stackrel{?}{=} g_t(x_0) h_t^{(1)}(r_1) h_t^{(2)}(r_2), \quad (13.1)$$

with  $r_1, r_2$  independent residual coordinates and  $h_t^{(1)}, h_t^{(2)}$  one-dimensional densities.

*Proof.* A factorisation of the form (13.1) would, in Fourier space, correspond to a fully separated characteristic function

$$\hat{q}_t^{\text{factor}}(\ell_0, \ell_1, \ell_2) = \mathcal{G}_0(\ell_0) \cdot \mathcal{B}_1(\ell_1) \cdot \mathcal{B}_2(\ell_2),$$

where  $\mathcal{G}_0$  is the site factor and  $\mathcal{B}_1, \mathcal{B}_2$  are the bond factors. This would require in particular the quadratic form in  $(\ell_0, \ell_1, \ell_2)$  to be diagonal after the rotation. But we have computed  $M$  explicitly, and  $M_{12} = -\alpha\Delta \neq 0$  whenever  $\alpha \neq 0$  and  $\Delta \neq 0$  (i.e.  $t \neq 0$ ). Therefore no such factorisation exists.  $\square$

*Interpretation* (The physical origin of  $-\alpha\Delta$ ). The cross-term  $M_{12} = -\alpha\Delta$  comes from the expansion  $k_1^2 = (\ell_1 - \alpha\ell_2)^2 = \ell_1^2 - 2\alpha\ell_1\ell_2 + \alpha^2\ell_2^2$ . It reflects the fact that the OU-injected noise  $\sqrt{\Delta}Z_1$  in coordinate  $X_1$  contributes to both the first innovation channel (via  $\ell_1 = k_1 + \alpha k_2$ ) and the coupling with the second innovation's dynamics (via the shift  $k_1 \rightarrow \ell_1 - \alpha\ell_2$ ). Markov locality survives (there is no  $\ell_0\ell_2$  coupling in  $M$ ), but the two consecutive innovations pick up a shared dependence on  $Z_1$ , the OU noise that lives at the shared vertex  $A_1$  of the chain.

### 13.2 Schur complement: effective coupling of the innovations

Marginalising out  $\ell_0$  in the Gaussian sector gives the effective covariance of  $(\ell_1, \ell_2)$  as the Schur complement

$$\widetilde{M} := M_{(1:2,1:2)} - M_{(1:2,0)} M_{00}^{-1} M_{(0,1:2)}. \quad (13.2)$$

**Computation of the Schur complement.** Using (12.16):

$$M_{(1:2,0)} = \begin{pmatrix} \alpha\beta^2\sigma_0^2 \\ 0 \end{pmatrix}, \quad M_{00} = \Delta + \beta^2\sigma_0^2 = v, \quad M_{(0,1:2)} = \begin{pmatrix} \alpha\beta^2\sigma_0^2 & 0 \end{pmatrix}. \quad (13.3)$$

So

$$M_{(1:2,0)} M_{00}^{-1} M_{(0,1:2)} = \frac{1}{v} \begin{pmatrix} \alpha^2\beta^4\sigma_0^4 & 0 \\ 0 & 0 \end{pmatrix}. \quad (13.4)$$

Therefore

$$\widetilde{M} = \begin{pmatrix} \Delta + \alpha^2\beta^2\sigma_0^2 - \frac{\alpha^2\beta^4\sigma_0^4}{v} & -\alpha\Delta \\ -\alpha\Delta & \Delta(1 + \alpha^2) \end{pmatrix}. \quad (13.5)$$

Simplify the (1,1)-entry:

$$\begin{aligned} \Delta + \alpha^2\beta^2\sigma_0^2 - \frac{\alpha^2\beta^4\sigma_0^4}{v} &= \Delta + \alpha^2\beta^2\sigma_0^2 \cdot \frac{v - \beta^2\sigma_0^2}{v} = \Delta + \alpha^2\beta^2\sigma_0^2 \cdot \frac{\Delta}{v} \\ &= \Delta \left( 1 + \frac{\alpha^2\beta^2\sigma_0^2}{v} \right) = \frac{\Delta(v + \alpha^2\beta^2\sigma_0^2)}{v} = \frac{\Delta(\Delta + \beta^2\sigma_0^2(1 + \alpha^2))}{v}, \end{aligned} \quad (13.6)$$

where we used  $v = \Delta + \beta^2\sigma_0^2$  twice.

$$\boxed{\widetilde{M} = \begin{pmatrix} \frac{\Delta(\Delta + \beta^2\sigma_0^2(1 + \alpha^2))}{v} & -\alpha\Delta \\ -\alpha\Delta & \Delta(1 + \alpha^2) \end{pmatrix}, \quad \widetilde{M}_{12} = -\alpha\Delta.} \quad (13.7)$$

*Remark.*  $\widetilde{M}_{12} = -\alpha\Delta$  is the same cross-term as  $M_{12}$ . The marginalisation of  $\ell_0$  does not change the (1,2) off-diagonal: the coupling between  $\ell_1$  and  $\ell_2$  is *purely* a consequence of the OU noise at  $A_1$ , not of any indirect path through the site variable  $A_0$ .

# Chapter 14

## Exact four-term representation

We now give an exact real-space representation of  $p_t(x_0, x_1, x_2)$  as a sum of four terms, each involving a bivariate normal CDF. This is the  $K = 2$  analogue of the two-term representation (5.15).

### 14.1 Setup

By the chain rule,  $p_t(x_0, x_1, x_2) = p_t(x_0, x_1) p_t(x_2 | x_0, x_1)$ . By Proposition 2.1,  $X_2 \perp X_0 | X_1$ , so

$$p_t(x_2 | x_0, x_1) = p_t(x_2 | x_1). \quad (14.1)$$

The *second-step* conditional is the analogue of the  $K = 1$  conditional but with  $A_1$  (a non-Gaussian variable) in the role of the latent state. This is where the analysis deviates from  $K = 1$ .

**Non-Gaussian posterior.** The analogue of the step “ $A_0 | X_0$  is Gaussian” at the second step is: what is the conditional distribution of  $A_1 | X_1 = x_1$ ? Using Bayes:

$$p_t(A_1 = a | X_1 = x_1) \propto p_t(X_1 = x_1 | A_1 = a) p_t(A_1 = a). \quad (14.2)$$

$p_t(X_1 = x_1 | A_1 = a) \propto \exp(-(x_1 - \beta a)^2 / (2\Delta))$  is Gaussian in  $a$ . The prior  $p(A_1)$  is the density of  $A_1 = \alpha A_0 + \varepsilon_1$ , i.e. a Gaussian-Laplace convolution. This prior on  $A_1$  is exactly of the Normal-Laplace type we analysed in Chapter 5, with appropriate parameters:

$$A_1 \sim \mathcal{N}(\alpha\mu_0, \alpha^2\sigma_0^2) * \text{Lap}(0, b). \quad (14.3)$$

The posterior  $A_1 | X_1 = x_1$  is therefore *not* Gaussian.

### 14.2 Proof sketch of the four-term representation

**Proposition 14.1** (Four-term representation of  $p_t(x_0, x_1, x_2)$ ). *There exist explicit functions  $\mu_{\sigma_1, \sigma_2}(x_0, x_1, x_2)$  and  $\Sigma_{\sigma_1, \sigma_2}$  for each of the four sign choices  $(\sigma_1, \sigma_2) \in \{+, -\}^2$ , such that*

$$p_t(x_0, x_1, x_2) = \sum_{\sigma_1, \sigma_2 \in \{\pm 1\}} C_{\sigma_1, \sigma_2}(x_0, x_1, x_2) \Phi_2(\mu_{\sigma_1, \sigma_2}(x); 0, \Sigma_{\sigma_1, \sigma_2}), \quad (14.4)$$

where  $\Phi_2(\cdot; 0, \Sigma)$  denotes the bivariate normal CDF with mean 0 and covariance  $\Sigma$ , and  $C_{\sigma_1, \sigma_2}$  is an explicit exponential prefactor depending on the sign pattern.

*Sketch.* Write the joint density in latent form as

$$p_t(x_0, x_1, x_2) = \iint p_t(x_0, x_1, x_2, u_1, u_2) du_1 du_2, \quad (14.5)$$

where  $u_i := \varepsilon_i$  and the integrand is the joint density with the two innovations explicit. The Laplace densities of  $u_1, u_2$  contribute  $|u_i|$  terms; splitting each by sign,

$$\text{Lap}(u; 0, b) = \sum_{\sigma \in \{\pm 1\}} \frac{1}{2b} \mathbf{1}_{\{\sigma u \geq 0\}} e^{-\sigma u/b}. \quad (14.6)$$

Over each quadrant  $\{\sigma_1 u_1 \geq 0, \sigma_2 u_2 \geq 0\}$ , the exponent in the integrand is a quadratic-plus-linear function of  $(u_1, u_2)$ . Completing the squares (the  $K = 2$  analogue of the  $K = 1$  step of Section 5.2) reduces each quadrant integral to a bivariate normal CDF  $\Phi_2$  with an explicit mean vector  $\mu_{\sigma_1, \sigma_2}(x)$  and covariance  $\Sigma_{\sigma_1, \sigma_2}$  – both depending parametrically on the constants  $(\alpha, \beta, \sigma_0, \mu_0, b, \Delta)$ . Collecting terms over the four sign choices gives (14.4).  $\square$

*Remark* (Why this is as far as we can go in closed form). The bivariate normal CDF  $\Phi_2$  is not elementary (no closed form in exponentials or rationals), but it is numerically stable via Genz’s algorithm and is treated as a standard special function. Hence the  $K = 2$  density is “as closed-form as one could reasonably hope”: a finite sum over sign patterns of elementary prefactors times a standard bivariate Gaussian CDF. At  $K = K$ , the analogous expression involves a sum of  $2^{K-1}$  terms, each with a  $(K - 1)$ -variate normal CDF. This exponential-in- $K$  scaling is the *analytical* obstruction to a full closed form at large  $K$ : it does not prevent numerical evaluation, but it does prevent a single tidy expression.

### 14.3 Practical numerical route

For simulations and visualisation, the four-term representation is less useful than direct 3D Fourier inversion of the characteristic function (12.7):

$$p_t(x_0, x_1, x_2) = \frac{1}{(2\pi)^3} \iiint e^{-i(k_0 x_0 + k_1 x_1 + k_2 x_2)} \hat{p}_t(k_0, k_1, k_2) dk_0 dk_1 dk_2. \quad (14.7)$$

The integrand decays as  $|k_i|^{-2}$  along the innovation frequencies and as Gaussian along all others, so a uniform grid truncation is fine for moderate tolerances. In practice we obtain 3–5% accuracy with  $N^3 = 56^3$  grid points. The  $K = 2$  numerical script in the bundle implements this and cross-checks it against a  $2 \cdot 10^6$ -sample Monte Carlo estimate.

## Chapter 15

# Conjectural rank structure of the $K=2$ curvature field

We do not have a full closed-form analogue of Theorem 7.1 for  $K = 2$ . But the numerical exploration and structural analysis suggest the following.

**Conjecture 15.1** (Rank- $\leq 2$  structure of  $H_t^{(K=2)}(x)$ ). For the Laplace/OU chain of Chapter 12, the curvature field  $H_t(x_0, x_1, x_2) = -\nabla^2 \log p_t(x_0, x_1, x_2)$  satisfies

$$H_t(x) = H_t^{\text{site}} + \kappa_t^{(1)}(r_1) M_\rho^{(1)} + \kappa_t^{(2)}(r_2) M_\rho^{(2)} + \mathcal{O}(\alpha\Delta/v), \quad (15.1)$$

where

- $H_t^{\text{site}}$  is a constant  $3 \times 3$  Gaussian term;
- each  $M_\rho^{(i)}$  is a rank-1 matrix supported on the nearest-neighbour pair  $(i, i+1)$ ;
- $\kappa_t^{(1)}(r_1), \kappa_t^{(2)}(r_2)$  are scalar curvature functions of two residual coordinates  $r_1, r_2$ ;
- the residual  $\mathcal{O}(\alpha\Delta/v)$  correction captures the obstruction cross-coupling  $M_{12} = -\alpha\Delta$ .

*Interpretation* (What this conjecture says). In words: the  $K = 1$  rank-1 result weakens to rank- $\leq 2$  at  $K = 2$ , as one might expect from two independent innovations. The low-rank structure is *local* (each rank-1 piece couples only two neighbours), which is the Markov-locality signature in real space. The deviation from exactness is controlled by the single parameter  $c = \alpha\Delta/v$  introduced in Section 3.3. If this conjecture holds, it suggests a general pattern: *at  $K = K_*$ , the curvature field lives on a rank- $\leq K_*$  manifold of matrices, with rank-1 blocks along the chain, and a controllable perturbation of order  $c^{K_*-1}$ .*

Testing this conjecture is on the roadmap for the next phase of the thesis.



## Part VI

# Interpretation and synthesis

## Chapter 16

# What Gaussianity buys, what Markovianity buys

The central theoretical goal of the thesis programme is to disentangle the roles of Gaussianity and of Markovianity in shaping the structure of the joint score. The Laplace AR(1) benchmark is the simplest context in which these two ingredients can be unmixed. We now summarise what the exact results in this compendium say on that question.

### 16.1 A brief restatement of the problem

In the Gaussian AR(1) benchmark (studied separately in [3]), the joint density  $p_t(x_0, \dots, x_{K-1})$  is Gaussian, the score  $S(x, t)$  is linear in  $x$ , and the curvature  $H_t = \Sigma_t^{-1}$  is a single matrix whose tridiagonal-mass fraction  $\rho_{\text{tri}}(t)$  measures the persistence of Markov locality. There are two sources of structure at play:

- **Gaussianity of the innovation and prior** makes the noisy law Gaussian, hence the score linear, hence the curvature a constant matrix. It is the reason  $\Sigma_t^{-1}$  alone describes everything.
- **Markovianity of the clean chain** determines the *sparsity pattern* of  $\Sigma_0^{-1}$  and, more generally, the *fill-in rate* of  $\Sigma_t^{-1}$  as diffusion progresses.

These are conceptually distinct. The first is a statement about the law; the second is a statement about the conditional independence graph. In the Gaussian case they combine seamlessly, and it is easy to conflate them. Breaking Gaussianity (by swapping in a Laplace innovation) forces them apart.

### 16.2 What Gaussianity alone gave us in $K=1$

In the Gaussian AR(1) benchmark at  $K = 1$ : the clean residual is Gaussian, the OU channel is Gaussian, so everything is Gaussian. The curvature  $\kappa_t^{(G)} = 1/\tau_G^2$  is a constant. The rank-1 structure (7.7) holds exactly with  $\kappa_t^{(G)}$  in place of  $\kappa_t(r)$ . In this case, the shape of the curvature matrix – i.e. the rank-1 structure  $M_\rho = nn^\top$  – is *entirely dictated by Markovianity*, not by Gaussianity. It reflects the graph structure of the chain: one bond between  $X_0$  and  $X_1$ , and the residual direction  $n \propto (\rho, -1)$  is perpendicular to the residual ray  $r = 0$ . *This is exactly the shape we find in the Laplace case, too*: the rank-1 matrix  $M_\rho$  is unchanged by the Laplace-vs-Gaussian choice.

*Interpretation* (Markov locality is what gives the rank-1 structure). The matrix  $M_\rho$  in Theorem 7.1 has rank 1 because the chain has only one bond at  $K = 1$ . This is a structural claim about

the graph, not about the innovation law. Whether the innovation is Gaussian or Laplace, the curvature field still has rank-1 local structure; what changes between the two is the scalar in front ( $\kappa_t^{(G)}$  constant vs.  $\kappa_t(r)$  nonconstant in the Laplace case).

## 16.3 What Gaussianity alone broke when we swapped it for Laplace

Moving from Gaussian to Laplace kept the rank-1 *shape* but broke the *constancy* of the scalar in front:

- The curvature  $\kappa_t(r)$  is now a function of the residual coordinate, concentrated at  $r = 0$  and vanishing in the tails.
- The score  $\psi_t(r)$  saturates at  $\pm 1/\tilde{b}$  in the tails rather than growing linearly.
- The posterior  $A_1 \mid X_1 = x_1$  is no longer Gaussian at  $K = 2$ , which is ultimately the reason  $M_{12} \neq 0$  there.

Notice that the first two effects are *local in  $r$*  and do not affect the chain’s Markov graph. The third effect, the  $K = 2$  obstruction, is a genuine new phenomenon: it says that non-Gaussian priors do not propagate cleanly through OU noising. The single-step clean analysis is preserved; the multi-step version is not.

## 16.4 What Markovianity kept intact in $K=2$

Even in the  $K = 2$  case, where Gaussianity fails us, the Markov structure *still* survives in three ways:

- The joint law is still exactly Markov (Proposition 2.1):  $X_2 \perp X_0 \mid X_1$ .
- The Gaussian sector  $M$  of Theorem 12.1 has a strict tridiagonal coupling pattern ( $M_{02} = 0$ ) in the rotated coordinates: there is no direct coupling between the site frequency  $\ell_0$  and the distant innovation  $\ell_2$ .
- The rational (non-Gaussian) factor is *exactly diagonal*: one factor per innovation, independent of all others.

The only surviving non-diagonality is the specific cross-term  $M_{12} = -\alpha\Delta$ , coming from the shared OU noise at the shared vertex  $A_1$ . Markov locality survives; only the particular kind of “full factorisation into independent bonds” fails.

*Interpretation* (The refined conclusion). Gaussianity is what gives clean factorisation into independent bonds at  $K = K$ ; without it, the bonds are diagonal in Fourier space but *share OU noise* through the shared vertices of the chain. The graph-theoretic Markov locality (no direct  $\ell_0\ell_2$  coupling) survives the loss of Gaussianity; what fails is the stronger statistical independence.

## 16.5 Consequences for the thesis research direction

These observations directly inform the architecture of a learned score network for dynamic data with temporally correlated frames:

1. **Nearest-neighbour coupling is structurally correct.** The exact benchmark shows that the right coupling pattern at all  $t$  is nearest-neighbour (tridiagonal in Fourier, at least

for AR(1) dynamics under OU). A score network whose attention pattern respects this locality should perform well for sequential data, independently of whether the innovation is Gaussian.

2. **The innovation-frequency axis is the right coordinate.** Writing the score in innovation coordinates  $(\ell_0, \ell_1, \ell_2, \dots)$  rather than the raw  $(k_0, k_1, k_2, \dots)$  isolates non-Gaussianity to a single axis. A score network that factorises along the residual axis – rather than along the raw frame axis – should track the structure of the exact benchmark.
3. **Non-Gaussianity is concentrated at finite  $t$ .** The centre-benchmark gap  $\delta\kappa_t$  scales as  $1/a$  at small  $t$  and as  $1/a^4$  at large  $t$ . Non-Gaussian fine-tuning is critical at early diffusion time; at late time the Gaussian benchmark is extremely accurate, which justifies Gaussian tails in the score network.
4. **The obstruction sets a complexity floor.** Exact closed-form solutions at  $K \geq 2$  require a finite sum of  $2^{K-1}$  multivariate Gaussian CDFs. This exponential-in- $K$  scaling is the analytic signature of why an exact score network is not feasible for realistic sequence lengths; it is also why parametric low-rank approximations – such as the conjectural rank- $\leq K$  structure of Conjecture 15.1 – are the right next step to investigate.

# Chapter 17

## Reading the heatmaps

We close with a pedagogical re-reading of the  $K = 1$  heatmaps shown in the main memo (Figure 2). The goal is to explain, step by step, what the geometry of the two-dimensional curvature field is telling us.

### 17.1 The residual ray as a one-dimensional manifold

The residual coordinate  $r = x_1 - \nu - \rho x_0$  vanishes on the line  $x_1 = \nu + \rho x_0$  in the  $(x_0, x_1)$  plane. This line is the *residual ray* – a one-dimensional manifold through the bivariate domain on which the Laplace cusp lives.

The slope of the ray is  $\rho$ , which interpolates between  $\alpha$  at  $t = 0$  (the clean AR(1) slope) and 0 as  $t \rightarrow \infty$  (when the noisy  $X_1$  decouples from  $X_0$ ).

### 17.2 Interpretation of $H_{11}(x)$

The entry  $H_{11}(x) = \kappa_t(r)$  depends on  $(x_0, x_1)$  only through the residual  $r$ . So  $H_{11}$  is constant along *any* line parallel to the residual ray, i.e. along the family of lines  $x_1 = \nu + \rho x_0 + r$ . Geometrically,  $H_{11}$  shows a “stripe” pattern: high values concentrated in a band around the residual ray, decaying to zero as one moves away.

**Why this step?** Why stripes? Because the non-Gaussianity lives entirely in the one residual direction. Anywhere you can move *parallel to* the ray, the residual  $r$  stays the same and so does  $\kappa_t(r)$ . You must move *perpendicular to* the ray to see  $\kappa_t(r)$  change.

### 17.3 Interpretation of $H_{01}(x)$

The off-diagonal entry is  $H_{01}(x) = -\rho \kappa_t(r)$ . It is a *multiple* of  $H_{11}$ , with the same stripe pattern and proportionality  $-\rho$ . A positive  $\rho$  (e.g.  $\alpha > 0$ ) gives a negative off-diagonal; a negative  $\rho$  gives a positive off-diagonal; and  $\rho = 0$  gives no off-diagonal at all. At  $t = 0$  where  $\rho = \alpha$ , the off-diagonal reflects the clean correlation. At  $t \rightarrow \infty$  where  $\rho \rightarrow 0$ , the off-diagonal vanishes – the two coordinates decouple.

### 17.4 What would change if we swapped the innovation law?

Under the exchange Laplace  $\rightarrow$  Gaussian: the heatmaps become *constant* in color, with the stripe pattern replaced by a single uniform value. The rank-1 structure is unchanged; only the scalar  $\kappa_t$  in front of it flips from a  $r$ -dependent Laplace curvature to a constant Gaussian curvature. This is the graphical embodiment of the interpretation of Section 16: Markov locality  $\rightarrow$  stripes; Gaussianity  $\rightarrow$  the color of each stripe is constant.

Under a general non-Gaussian innovation: the stripe pattern persists (Markov still holds), but the *shape* of  $\kappa_t(r)$  as a function of  $r$  changes. For multimodal innovations,  $\kappa_t$  can even be negative, indicating regions of the residual-ray direction where the density is locally convex in log-space (“anti-confining” regions). Investigating this is listed as one of the open questions in the main memo.

# Chapter 18

## Open questions revisited

The main memo lists five open questions. Here we indicate, for each, what specific missing piece of the exact picture it corresponds to and how it might be attacked.

1. **Centre-tail gap scaling for general innovation laws.** From the ODE (8.11),  $h_t''(0) = (h_t(0) - \phi_\tau(0))/\tilde{b}^2$ . Thus  $\kappa_t(0) = -h_t''(0)/h_t(0) = -(1 - \phi_\tau(0)/h_t(0))/\tilde{b}^2$ . This is a universal formula: for *any* innovation law with a density that we can evaluate at 0, the centre curvature is determined by  $h_t(0)$ . So the centre-tail gap  $\kappa_t(0) - \kappa_t^{(G)}$  reduces to studying the functional  $\varepsilon \mapsto h_t(0)$ . For a variance-matched Gaussian, this ratio is always  $\sqrt{2/\pi}/a$  at leading order at small  $a$ , with model-dependent subleading corrections controlled by higher cumulants. A rigorous treatment is accessible via Edgeworth expansion.
2. **Variance-Gamma bridge.** The VG family  $\hat{f}_\varepsilon(\ell) = (1 + b^2\ell^2/(2\nu))^{-\nu}$  interpolates Laplace ( $\nu = 1$ ) to Gaussian ( $\nu \rightarrow \infty$ ). At integer  $\nu$ , the residual density  $h_t^{(\nu)}$  is a sum of  $\nu$  Laplace-type terms (via partial fractions), and the centre curvature admits a closed form as a generalisation of (9.2). Computing this and tracking  $\kappa_t^{(\nu)}(0)$  along the VG family would quantify how the rank-1 scalar  $\kappa_t$  turns on with non-Gaussianity.
3. **Multimodal innovations.** For the symmetric Gaussian-mixture innovation  $\varepsilon \sim \frac{1}{2}\mathcal{N}(-m, \sigma^2) + \frac{1}{2}\mathcal{N}(+m, \sigma^2)$ , the characteristic function is  $\hat{f}_\varepsilon(\ell) = \cos(m\ell)e^{-\sigma^2\ell^2/2}$ , and the residual  $R_t = \beta\varepsilon + G_t$  has density

$$h_t(r) = \frac{1}{2}[\phi_{st}(r - \beta m) + \phi_{st}(r + \beta m)], \quad s_t^2 = \tau^2 + \beta^2\sigma^2. \quad (18.1)$$

For small  $t$ ,  $h_t$  has two peaks at  $\pm\beta m$ ; for large  $t$ , they merge.  $\kappa_t(r)$  can be *negative* in a range of  $r$ . The critical diffusion time  $t^*$  at which the bimodality disappears is an explicit function of  $m/\sigma$ .

4. **Nonlinear locality observables.** Define  $T_d(x, t) := \sum_{|i-j| \leq d} H_t(x)_{ij}^2 / \|H_t(x)\|_F^2$ , the fractional Frobenius mass supported within a band of width  $d$ . In the Gaussian benchmark,  $T_d(x, t)$  is constant in  $x$  and matches the Gaussian tridiagonal-mass measure  $\rho_{\text{tri}}(t)$ . In the Laplace benchmark,  $T_d(x, t)$  depends on  $x$  through  $\kappa_t(r)$ ; its pointwise expectation is  $\bar{T}_d(t) = \mathbb{E}_{X \sim p_t}[T_d(X, t)]$ , a true functional replacement for the Gaussian measure. Computing  $\bar{T}_d$  as a function of  $t$  and  $d$  is the bridge between the Gaussian benchmark and the non-Gaussian benchmark.
5. **Testing Conjecture 15.1.** Systematic evaluation of  $H_t$  on a 3D grid of  $(x_0, x_1, x_2)$  values (say  $30^3 = 27000$  points), SVD of each  $3 \times 3$  matrix, statistical analysis of the resulting rank distributions. The conjecture predicts that the rank- $\leq 2$  approximation error decays as  $(\alpha\Delta/v)^2$ . Verifying this numerically is a direct test.

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